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Apparent polar wander and reversal stratigraphy of the Palaeo–Mesoproterozoic southeastern McArthur Basin, Australia

M. Idnurm, J.W. Giddings, K.A. Plumb

Australian Geological Survey Organisation, G.P.O. Box 378, Canberra, A.C.T. 2601, Australia

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Abstract

Approximately 1800 samples from volcanic, clastic and carbonate sequences of the southeastern McArthur Basin in northern Australia were analysed palaeomagnetically to define in detail the Australian apparent polar wander path (APWP) for a period centred at ~1670 Ma and to develop a Proterozoic reversal stratigraphy for correlation and dating. Twelve palaeomagnetic poles are interpreted as primary, including two that are preliminary. These increase the Australian palaeomagnetic database for the Proterozoic by ~30%. Another ten poles record three, or possibly four, periods of overprinting. The study also yields one of the oldest reversal records reported so far.

The ten better-established primary poles were obtained from the lower three of the four main stratigraphic subdivisions of the Basin: the Tawallah, McArthur and Nathan Groups. These poles define an angular trajectory of about 120° for a time span of at least 75 m.y., and possibly exceeding 100 m.y. They confirm earlier preliminary results that suggested a major revision of the Proterozoic APWP for Australia. The two preliminary poles were obtained from the youngest sub-division of the Basin, the Roper Group, and confirm a large age gap between the Nathan and Roper Groups.

The poles for the oldest overprints were from igneous units in the Tawallah Group and appear to record metasomatism at the end of the last major period of volcanism in the basin. The pole for the next overprint falls on the path near the apex of a hairpin bend; it coincides with a likely period of rifting and high heat flow that accompanied the deposition of the upper McArthur Group. The third overprint, apparently acquired during the long time-break separating the Nathan and Roper Groups, was found mostly in red dolomitic rocks.

The geomagnetic reversal record from this study represents an accumulated stratigraphic thickness of 2100 m, excluding gaps where the polarity remained undetermined because of overprinting, unstable remanences, intervals not sampled, or other causes. Reversal patterns from duplicate sections at different localities were compared for two formations, and were found to be consistent. Several parts of the reversal column appear to be distinctive enough for use in stratigraphic correlation.

1. Introduction

The Precambrian APWP for Australia has been determined mainly from remanence directions of dated igneous units. Such directions, while essential for age calibration, are likely to leave large gaps because of the intermittent and sporadic nature of igneous activity. Sedimentary sequences generally provide more continuous geologic records, but their ages are more difficult to establish. Nevertheless, for polar wander studies such sequences, if they have even the broadest age control, can supplement igneous units by defining segments of trajectories between dated

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poles. Moreover, the stratigraphic order of remanence directions in a continuous sequence may supply the direction of pole movement along the path. This is especially important in the Precambrian where path segments may have to be fitted to a few, widely spaced poles, leaving the direction of polar wander doubtful.

Besides their use in polar wander studies, sedimentary sequences may also provide a record of geomagnetic reversals with applications to correlation and dating of rock units. So far, only very few attempts to set up a reversal stratigraphy for correlation have been reported for the Precambrian (Evans et al., 1975; Bingham and Evans, 1976; Elston and Bressler, 1980; Elston, 1984; Thomas and Piper, 1992). This is partly because of the difficulty of separating and identifying primary remanences in partially overprinted rocks and partly because of the lack of suitable continuous sequences. In Australia work has been limited to a narrow interval spanning the Precambrian–Cambrian boundary (Kirschvink, 1978; Burek et al., 1979; Chamalaun et al., 1989).

This paper describes a polar wander and polarity reversal study of a part of the Palaeo–Mesoproterozoic McArthur Basin of northern Australia (Fig. 1). The basin comprises sediments, and minor volcanics and intrusives, which together form the platform cover on a Palaeoproterozoic basement. The sequences appear to be unmetamorphosed, but have in some regions been intensely metasomatized. Lateritic ferruginisation is generally restricted to flat areas. Except near faults, deformation is normally by gentle folding.

The McArthur Basin covers an area of 200,000 km^2 and has strata of variable thicknesses up to at least 12 km. The present study concerns its southeastern part (Figs. 1 and 2) which contains both clastics and carbonates. Most rocks of this region were formed during a $\sim$100 m.y. period centred around 1670 Ma for which the Australian APWP has few dated poles. At some localities the outcrop is sufficiently continuous and extensive for polarity reversal studies.

Preliminary results of this work, based on relatively few samples, have been described earlier (Plumb, 1985; Idnurm and Giddings, 1988). Although the preliminary pole positions are revised, the pole trajectory needs little change. However, several remanences that were tentatively assumed to be primary now turn out to be overprints, leading to considerable modifications in the polarity pattern. A related study of thick sequences of predominantly quartz sandstones and conglomerates of the Kombolgie Formation in the northwestern McArthur Basin will be reported separately.

1.1. Geology

The McArthur Basin is part of the North Australian Craton (Plumb, 1979), which extends from the Kimberley Block in the west to the Mt Isa Inlier in the east, and is bounded by the Arunta and Pilbara Blocks to the south (Fig. 1). The basement rocks of the craton are Palaeoproterozoic and Archaean. The overlying platform cover comprises five main Palaeo–Mesoproterozoic basins, including the McArthur Basin, whose inter-relationships are discussed by Plumb et al. (1990).

Recent comprehensive descriptions of the southeastern McArthur Basin geology, by Jackson et al. (1987) and Plumb et al. (1990), provide the basis for the following summary. The basin at the southeast consists of a 50 km wide, north-trending rift, the Batten Trough (Fig. 1) with deposits up to 10 km thick, and flanking areas with deposit thicknesses of up to 4 km. The Batten Trough was a subsiding half-graben throughout much of the basin history. It is dissected by numerous northerly and northwesterly faults, many of which were active at the time of deposition. Later, complex block faulting inverted the rift and further elevated synrift tilt-blocks. The shallower areas flanking the Trough are interpreted as former depositional shelves.

The sedimentary succession is subdivided into four groups, shown in Fig. 3 (Jackson et al., 1987). The oldest, the Tawallah Group, lies unconformably on an Orosirian (middle Palaeoproterozoic) basement, and consists mainly of quartz arenites and basic to intermediate volcanics. At several localities the top members of this group contain rhyolites, or an apparently

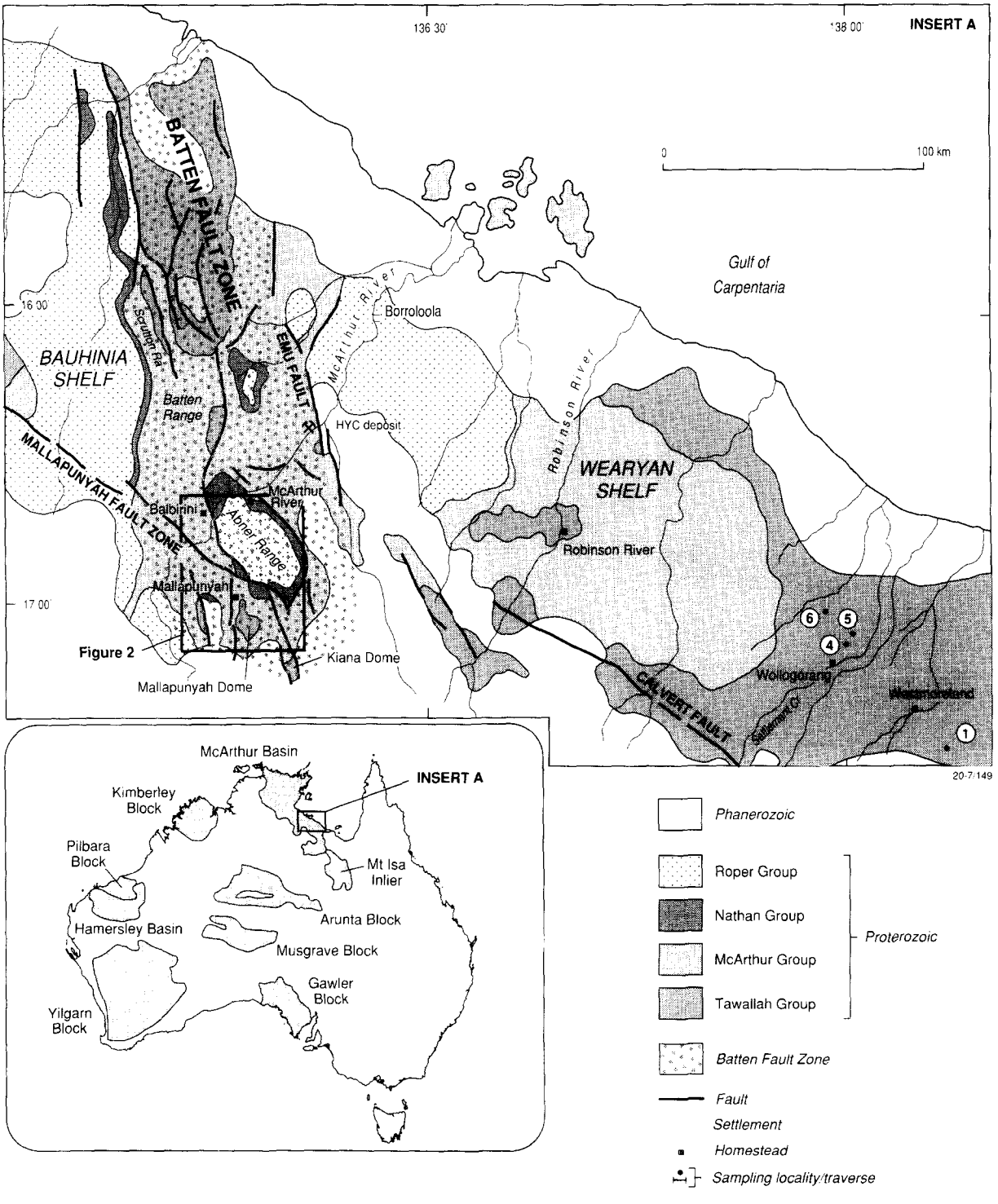


Fig. 1. Simplified geological map of the southeastern McArthur Basin showing main structural elements and sampling localities (after Jackson et al., 1987). Map of Australia shows the McArthur Basin relative to other major Precambrian tectonic units.

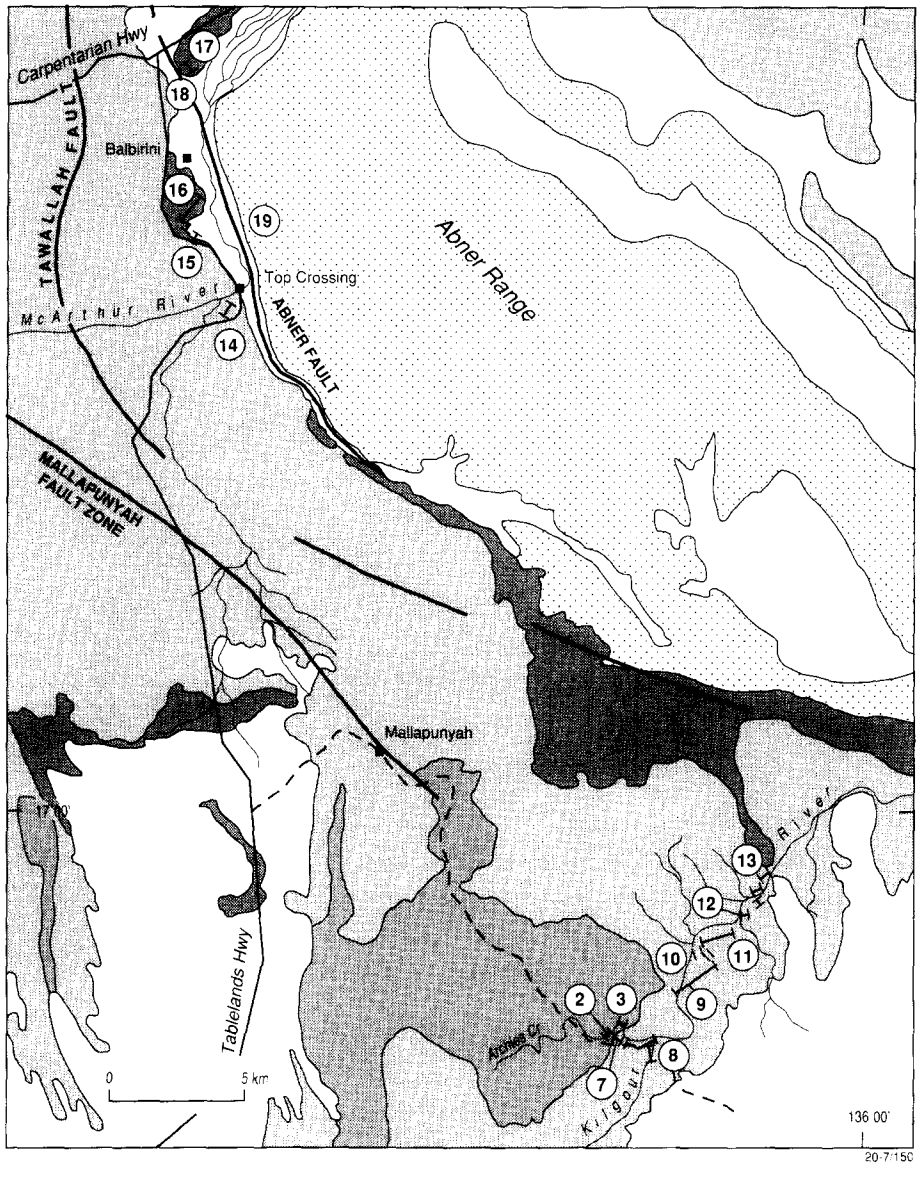


Fig. 2. Simplified geology of the main sampling area. Key as for Fig. 1.

comagmatic microgranite. Most volcanics have been severely altered. Plumb et al. (1990) suggest that the Tawallah Group was deposited in a variety of shallow-marine and continental environments extending across a series of synrift tilt-blocks, all within a broad extensional terrain.

The McArthur Group unconformably overlies the Tawallah Group, and is in turn unconformably overlain by the Nathan Group. The lithologies of the two younger groups are similar: both comprise mainly dolomites and dolomitic siltstones and shales, except the lowermost McArthur Group in which sandstones are common. Stromatolites and hypersalinity indicators (e.g., pseudomorphs after evaporites) are abundant throughout. All units contain evidence of a very shallow water origin, including diastems and disconformities. Diagenetic/authigenetic alteration is widespread, particularly in the upper McArthur Group (e.g., Barney Creek Forma-

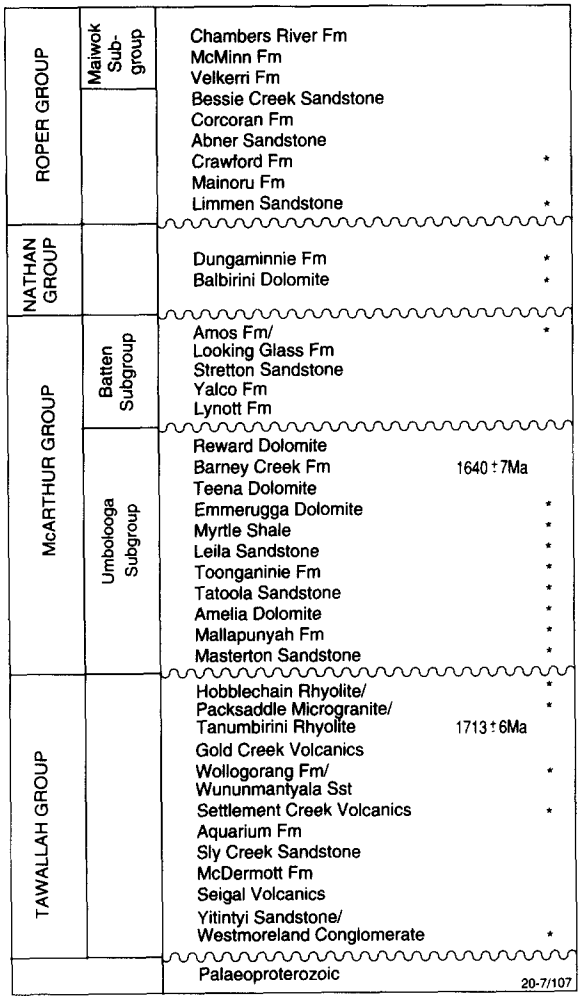


Fig. 3. Main Palaeo–Mesoproterozoic stratigraphic units and principal unconformities of the southeastern McArthur Basin. Formations sampled in the present study are indicated by asterisks. Dates are from U–Pb zircon microprobe analyses.

tion). The lower McArthur Group (Umbolooga Subgroup) contains peritidal to continental carbonate and sabkha facies sediments that were deposited on a broad shelf within the confines of the Batten Trough half-graben. Rifting and penecontemporaneous faulting became more pronounced during the deposition of the topmost Umbolooga Subgroup and the overlying Batten Subgroup. The deposition of these units, in hypersaline and alkaline lake complexes, was accompanied by pervasive diagenetic/authigenetic alteration in a regime of high heat flow and by deposition of felsic tuffs. The major stratiform HYC Pb–Zn–Ag deposit (Logan et al., 1990) was formed within the Barney Creek Formation during this interval of tectonic instability.

Following basin-wide uplift, erosion, and the formation of a regional regolith, cyclic carbonate complexes of the unconformably overlying Nathan Group were deposited in hypersaline playas and lakes or on peritidal flats. These were located in a broad sag basin extending far beyond the boundaries of the Batten Trough.

The youngest sub-division of the Basin, the Roper Group, is separated from the Nathan Group by a fourth regional unconformity. In contrast to predominantly carbonates in the underlying successions, the Roper Group comprises alternating sequences of quartz arenites and siltstones/shales. These were deposited in another sag basin and are remarkably uniform laterally. Environments include a variety of marginal, shallow- and deeper-marine settings, reflecting alternating basin-wide transgressions and regressions. The Roper Group is currently being explored for hydrocarbons.

1.2. Geochronology

Several dates have been obtained for the southeastern McArthur Basin but, due to isotopic resetting, most provide only minimum ages.

A conventional U–Pb zircon date of 1725±40 Ma (Pietsch et al., 1991)[1] and an ion microprobe date of 1713±6 Ma (R.W. Page, in Haines et al., 1993) have been obtained recently from the Tanumbirini Rhyolite, at the top of the Tawallah Group. In comparison, earlier Rb–Sr age estimates for the Tawallah Group are appreciably younger: i.e., ~1556 Ma for the Aquarium Formation (McDougall et al., 1965), and 1488 to 1542 Ma for the Packsaddle Microgranite and Hobblechain Rhyolite (Tanumbirini Rhyolite equivalents) (McDougall et al., 1965; A. Webb quoted in Kralik, 1982). Indeed, the Rb–Sr estimates are considerably younger than even a U–

[1]All dates reported here have been adjusted for the standard decay constants listed by Steiger and Jäger, 1977. Precisions are given as two standard deviations.

Pb zircon date from the stratigraphically higher Barney Creek Formation (see below). The low values may be due to isotopic resetting by potassium metasomatism reported by Knutson et al. (1979), or to the process (discussed below) that has reset the ages of illite and kaolinite in the McArthur Group tuffs.

Recently, an ion microprobe U–Pb zircon date of 1640±7 Ma was obtained for a tuff layer in the Barney Creek Formation at the middle McArthur Group (R.W. Page, in Haines et al., 1993). This replaces the earlier conventional U–Pb zircon date of 1690 +29/−25 Ma (Page, 1981) which was apparently biased by inherited zircons, and provides the only age constraint within the McArthur Group. Plumb et al. (1990) suggest that the top of the McArthur Group may be older than 1610 Ma, a suggestion based on deformation at 1610 Ma of a stratigraphically equivalent group in the Mount Isa Inlier. Rb–Sr ages for the tuff are again much younger: 1537±52 and 1589±14 Ma, obtained on diagenetic illite and feldspar from the Barney Creek Formation tuff bed (Kralik, 1982). The cause of the age discrepancy is not clear. Jackson et al. (1987) suggest the possibility of isotopic resetting by contemporaneous metamorphism in the adjacent Mount Isa region. However, reflectances of hydrocarbon residues in drillcores indicate that the McArthur Group is sub- to overmature, suggesting that any metamorphism has been below the greenschist facies (Cook, 1982; Crick et al., 1988) and therefore too mild for thermal isotopic resetting. Alternatively, Plumb (1989) attributes total alteration of feldspar in the overlying Batten Subgroup tuffs, and by analogy in the Barney Creek Formation as well, to pervasive diagenesis/authigenesis of uncertain age.

Diagenetic ages have also been obtained by the Rb–Sr method for two members of the Roper Group: 1361±20 Ma determined for glauconites in the Crawford Formation, in the lower part of the Group (McDougall et al., 1965); and 1429±31 Ma for illites in the McMinn Formation from near the top (Kralik, 1982). The latter is the more reliable minimum estimate for the age of the Group. Comparison with the South Nicholson Basin, which adjoins the McArthur Basin to the southeast, suggests that the base of the Group is probably younger than the 1610 Ma folding at Mount Isa (Plumb et al., 1990).

The age of the Nathan Group can be constrained only between those of the McArthur and Roper Groups: by inference, probably between 1600–1500 Ma (Plumb, 1990), or a little older.

2. Sampling procedures and measurements

About 1800 samples were collected. The emphasis was on the upper Tawallah, McArthur, and Nathan Groups, because of their generally good exposures and stratigraphic continuities. Fig. 1 shows the sampling localities. Most are in the Batten Trough where the McArthur and Nathan Groups have their maximum known thicknesses and where streams have incised an uplifted terrain to expose the strata. The majority of samples was collected along two traverses. The first, a composite of several stratigraphic sections along the Kilgour River and its tributaries, spans a 900 m thickness of the upper Tawallah to middle McArthur Groups. The average stratigraphic interval for sampling was 1 m. The second traverse, on the northwestern flanks of the Abner Range, spans a 900 m thickness of the Nathan Group and was sampled at an average interval of 2.5 m. Several units were sampled elsewhere in the fault zone or near the eastern margin of the basin. The traverses are shown in Fig. 2 and, in greater detail, on the geologic map accompanying Jackson et al. (1987).

Most sampled sections had been measured previously by Abney level and Jacobs rod for sedimentological studies, many being the type sections of the respective formations. Additional measurements were needed only to interpolate between the existing stratigraphic levels or to establish traverse lines more suitable for palaeomagnetism (e.g., less likely to have experienced block rotations).

Most samples were cores obtained with a small portable drill; the rest were blocks. Sample orientation was normally by both a sun and a magnetic compass. The core surfaces were sand-

blasted lightly in the laboratory to remove any magnetic contaminants from drilling.

Approximately 2000 specimens were demagnetised stepwise at the Black Mountain Palaeomagnetic Laboratory in Canberra. The remanences were measured principally on a 2-axis ScT cryogenic magnetometer, with specimen inversion to obtain eight readings for each measurement. Demagnetization was by standard alternating field (AF) and thermal techniques: the former using either a Schonstedt 2-axis GSD-5 demagnetizer or a 2-axis instrument described by Senanayake (1981); the latter in a Schonstedt TSD-1 demagnetizer or in large-volume furnaces described by McElhinny et al. (1971). In preliminary work, thermal demagnetization gave usually less scattered orthogonal projection plots than AF demagnetization, and was therefore adopted for most subsequent measurements. However, for several sites thermal demagnetization alone gave large, directional scatters between samples. For these sites, samples were partially demagnetized by AF at peak intensities up to 100 mT before thermal treatment—a procedure that has successfully removed lightning-induced overprints (Cox, 1961; Graham, 1961; Purucker, 1974). Magnetic susceptibilities were monitored between heatings using a Collinson and Molyneux (1967) type instrument.

Remanence components were determined interactively by principal components analysis (Kirschvink, 1980). Generally, if the number of directions for a pole exceeded 20, the criterion for inclusion of a direction was set arbitrarily at a mean angular deviation (MAD) of $<5°$; if this number was less than 20 the angle was $<7°$. The formation mean directions were calculated from sample averages if more than one remanence direction was available; otherwise the specimen direction was used. Directions deviating by more than 60° from the mean were excluded from the final analysis.

Polarity assignment for some parts of the reversal column was complicated by the destruction of primary remanence by magnetic overprinting. However, even though the remanence directions could not be determined, the polarity was sometimes discernible. The reversal diagrams show the inclination and declination of the primary directions, if available. Polarity-only data are indicated by square symbols.

3. Sampling localities and results

The results are described in stratigraphic order from the oldest to the youngest. Normally only components that survived 200°C or 20 mT peak fields are mentioned; less stable components are assumed to be recent. If more than one specimen was measured from a sample, the stereo plot shows the sample mean.

Table 1 lists the sampling details, including the locality numbers referred to in the text, and Table 2 summarises the results. The stratigraphic sections in Table 1 are described by Jackson et al. (1987). Type sections are labelled with an asterisk in the Traverse column. Formations that did not yield useful results are included in the tables, but not discussed in the text.

3.1. Settlement Creek Volcanics

Ten dark-grey mafic samples were collected from the upper part of the Volcanics (Locality 2). Bedding was not evident at the site, and the site tilt correction was estimated from nearby interbedded sediments and the Wollogorang Formation. Because of large variations in bedding dips, this estimate is approximate.

Most specimens were demagnetized by AF, others thermally, revealing two components, I (intermediate) and H (high), distinguished by differences in coercivity and unblocking temperatures. The I component, which dominated the remanence, was removed at 15–100 mT or just below 600°C. It yielded a scattered group of WNW directions with negative inclinations (Fig. 4c) and a bedding-corrected mean 291.1°, −25.6° ($\alpha_{95}=16°$). Jackson et al. (1987) note that the Settlement Creek Volcanics are hydrothermally altered. This was confirmed in thin sections cut from palaeomagnetic samples. Therefore, component I is probably secondary. Its pole position on the APWP (SCVO in Fig. 26) suggests that the magnetization was ac-

Table 1
Sampling localities

Rock unit/ locality	Geogr. co-ord.[a]		Map reference[a] (1:100,000 maps)		Traverse[b]	No. of samples	Thickness (m)
	long. (°E)	lat. (°S)					
Westmoreland Conglomerate[e] Loc. 1	138.34	17.48	Westmoreland	179650	–	41	~100
Settlement Creek Volcanics Loc. 2	135.89	17.09	Kilgour	950103	–	10	–
Wollogorang Formation Loc. 3	135.90	17.08	Kilgour	955110	K78/02	34	70
Hobblechain Rhyolite Loc. 4	137.99	17.13	Wollogorang	174033	–	26	~30
Packsaddle Microgranite							
Loc. 5	137.99	17.10	Wollogorang	186068	–	9	–
Loc. 6	137.92	17.05	Wollogorang	110130	–	10	–
Masterton Sandstone							
Loc. 7	135.90	17.09	Kilgour	956103	K78/01[d]	71	50
Mallapunyah Formation							
Loc. 8	135.91	17.09	Kilgour	960100	K77/11, 12[d]	358	220
Loc. 9	135.92	17.07	Kilgour	978128	K77/10	62	~90
Amelia Dolomite							
Loc. 8	135.91	17.09	Kilgour	967098	K77/12	41	35
Loc. 9	135.92	17.05	Kilgour	985138	K77/10[d]	192	170
Tatoola Sandstone							
Loc. 10	135.92	17.05	Kilgour	985145	K77/7, 9[d]	72	125
Tooganinie Formation							
Loc. 10, 11	135.93	17.05	Kilgour	990150	K77/3[d], 7[d]	163	244
Leila Sandstone[e]							
Loc. 12	135.94	17.03	Kilgour	101160	K77/8	2	5
Myrtle Shale							
Loc. 12	135.94	17.03	Kilgour	102160	K77/8[d]	41	56
Emmerugga Dolomite							
Loc. 13	135.94	17.03	Kilgour	005172	K77/1, 4, 5	196	157
Loc. 14	135.76	16.80	Kilgour	805430	4[c,d]	90	230
Amos Formation[e]							
Loc. 15	137.74	16.76	Kilgour	789466	M77/3[d]	37	87

Table 1 (continued)

Rock unit/ locality	Geogr. co-ord.[a]		Map reference[a] (1:100.000 maps)		Traverse[b]	No. of samples	Thickness (m)
	long. (°E)	lat. (°S)					
Balbirini Dolomite							
Loc. 16	135.74	16.75	Kilgour	788777	M77/4, 5[d]	257	897
Dungaminnie Formation[e]							
Loc. 17	135.74	16.69	Kilgour	784543	M77/6[d]	28	92
Roper Group							
Loc. 18	135.74	16.70	Kilgour	788540	–	8	2
Loc. 19	135.76	16.77	Kilgour	810460	–	24	~50

[a] Where the samples were collected along a traverse line these co-ordinates indicate the locality centre.
[b] Jackson et al., 1987.
[c] Section 4, Plate 4, Brown et al., 1969.
[d] Indicates type section for the formation.
[e] Formation did not give a useful result because of unstable remanence or large directional scatter.

quired at or before the McArthur Group deposition had commenced. The harder H component remained unresolved because of directional instability (Figs. 4a and 4b).

3.2. Wollogorang Formation

This formation was sampled at Locality 3. The sequence comprises dark-grey, pale-brown and red-brown shales and dololutites, interpreted as lacustrine/fluvial deposits. The bedding dip and strike change considerably at the beginning of the section near a prominent fault, providing a palaeomagnetic fold test. Elsewhere the bedding varies irregularly at the metre scale, and masks a regional dip seen in adjacent units. Thirty-four samples were collected at approximately 3 m intervals over a 70 m stratigraphic thickness.

The basal 40 m of the sequence contained a one-polarity, single-component remanence, I, that unblocked usually at 400–550°C. Above 40 m, an additional component, H, of slightly higher unblocking temperature and two polarities appeared (Figs. 5a and 5b). Because of scatter, the direction of H could be defined in only 11 out of the 26 samples, and sometimes only by anchoring the principal component linear segment to the origin. However, in most samples its polarity was evident in the orthogonal projection plots.

The remanence directions before and after bedding corrections are shown in Figs. 5c and 5d). Bedding correction increased the directional scatter of the I component, for which the fold test of McFadden (1990) was negative at the 99% confidence level (Table 3). Thus I is an overprint. On the other hand, bedding correction reduced the directional scatter of H. McFadden's fold test for H was inconclusive, but McElhinny's (1964) test showed the reduction to be statistically significant at 95% confidence level (Table 3), suggesting that H was acquired before deformation.

As discussed later, regional tilting appears to post-date the I component. Adoption of a tilt correction from the nearby Masterton Formation (14° towards 145°) gives the mean direction of I as 354.9°, −39.0° (α_{95}=5.6°). The mean for the bedding-corrected H component, after inversion of the positive-inclination directions, is 284.8°, −22.8° (α_{95}=8.5°). The reversal pattern for the sequence is shown in Fig. 6.

3.3. Hobblechain Rhyolite

The Hobblechain Rhyolite and Packsaddle Microgranite (discussed below), topmost mem-

Table 2
Summary of palaeomagnetic results

Rock unit/ locality[a]	Comp.[b]/ colour	No. of samples (specimens)	Bedding corr.	Fold test[c]	NRM int.[d] (mA m^{-1})	Remanence direction				Pole					Palaeo-lat. (°)
						decl. (°)	incl. (°)	k	α_{95} (°)	lat. (°S)	long. (°E)	K	A_{95} (°)	Acr.	
Settlement	I(O?)	8	No		10–2000	291.3	−40.6	12.7	16.1	26.6	205.8	13.5	15.6		28.2
Creek Volcs.		(17)	Yes			*291.1*	*−25.6*	*12.7*	*16.1*	*24.3*	*217.4*	*18.5*	*13.2*	SCVO	*14.5*
	H	–	–			Presence inferred, direction undetermined									
Wollogorang	I(O)	27	No		0.5–10	2.7	−50.7	25.9	5.6	74.1	126.5	20.4	6.3		32.8
Formation		(30)	Yes	F−		342.7	−23.6	24.5	9.1						
			Yes[e]			*354.9*	*−39.0*	*25.9*	*5.6*	*83.1*	*179.3*	*20.4*	*6.3*	WFKO	
	H	11	No			271.5	−50.0	11.6	14.0	10.9	192.0	8.9	16.2		35.4
		(16)	Yes	F+		*284.8*	*−22.8*	*29.6*	*8.5*	*17.9*	*218.2*	*41.8*	*7.2*	WFK	*12.2*
Hobblechain	I1	8	No		5–50 (L)	113.6	32.2	17.3	13.7	27.1	215.7	26.7	10.9		18.4
Rhyolite		(16)	Yes		35–3000 (U)	*103.1*	*37.1*	*17.3*	*13.7*	*17.6*	*209.9*	*22.6*	*11.9*	HR	*21.8*
	I2(O)	–	–			Undetermined									
	H(O)	17	No			115.3	42.5	11.3	11.1	29.4	208.4	18.3	8.6		25.1
		(26)	Yes			*100.0*	*47.2*	*15.6*	*11.1*	*16.4*	*202.0*	*18.0*	*8.7*	HRO	*29.0*
Packsaddle Microgranite					10–700										
Loc. 5	I	–	–			Large directional scatter									
	H(O)	6	No			106.3	54.6	14.3	18.3	24.1	194.4	9.2	23.3		37.0
		(6)	Yes			*83.2*	*55.7*	*14.4*	*18.3*	*6.2*	*189.4*	*9.9*	*22.4*		*38.6*
Loc. 6	I	–	–			Large directional scatter									
	H(O)	9	No			115.7	43.8	41.8	8.1	30.0	207.0	48.9	7.4		26.3
		(27)	Yes			*102.9*	*47.1*	*41.8*	*8.1*	*18.9*	*202.5*	*41.6*	*8.1*		*28.9*
Loc. 5+6	H(O)	15	No			112.5	48.1	22.6	8.2	27.9	202.0	17.8	9.3		30.4
(combined)		(33)	Yes			*96.1*	*50.8*	*21.3*	*8.5*	*14.1*	*197.3*	*16.2*	*9.8*	PMGO	*31.9*
Masterton	H	18	No		0.5–20	287.1	−16.2	19.6	8.0	18.8	222.4	26.3	6.9		8.7
Sandstone		(18)	Yes			*288.2*	*−5.3*	*19.3*	*8.1*	*18.2*	*228.5*	*27.7*	*6.7*	MAS	*2.9*
Loc. 7															
	O	14	No			*5.4*	*−31.1*	*24.4*	*8.2*	*85.3*	*46.0*	*28.6*	*7.6*	MASO	*17.1*
		(14)	Yes			0.7	−25.3	23.7	8.3	86.4	321.5	27.0	7.8		13.5
Mallapunyah Formation					0.5–5										
Loc. 8	O	15	No			353.4	−41.5	14.3	8.9	80.7	174.0	15.2	10.2		24.4
		(15)	Yes			*351.3*	*−38.2*	*19.6*	*8.9*	*80.7*	*193.4*	*16.3*	*9.8*	MALO	*21.9*
Loc. 8	H	114	No			120.4	43.4	15.9	3.4	34.1	204.6	13.7	3.7		26.9
		(119)	Yes			*121.5*	*32.8*	*17.5*	*3.3*	*34.5*	*214.3*	*18.2*	*3.2*		*18.9*
Loc. 9	H	10	No			140.1	44.4	20.9	11.0	51.9	204.6	14.4	13.2		26.5
		(10)	Yes			*127.2*	*35.4*	*22.0*	*10.5*	*40.6*	*214.0*	*19.6*	*11.2*		*19.9*
Loc. 8+9	H	124	No			122.0	43.6	15.7	3.3	35.5	204.6	13.3	3.6		27.0
		(129)	Yes	F+		*122.0*	*33.0*	*17.8*	*3.1*	*35.0*	*214.3*	*18.3*	*3.1*	MAL	*19.0*
Amelia Dolomite					0.2–23										
Loc. 8	O1	21	No			356.5	−41.9	19.2	7.5	81.8	156.6	16.6	8.0		24.8
		(22)	Yes			*350.2*	*−40.9*	*23.3*	*6.7*	*78.7*	*184.0*	*22.2*	*6.9*	AMEO1	*24.4*
Loc. 9+10	O2	58	No			*242.8*	*51.7*	*27.9*	*3.6*	*32.2*	*74.8*	*18.7*	*4.4*	AMEO2	*33.2*
		(62)	Yes			247.7	65.4	26.8	3.7	27.9	92.5	13.4	5.3		48.7
Loc. 9+10	Grey	29	No			336.0	−56.2	13.7	7.5	59.6	174.3	9.2	9.4		39.3
		(35)	Yes			*314.5*	*−58.8*	*13.9*	*7.4*	*43.5*	*184.2*	*9.2*	*9.3*		*41.6*

Table 2 (continued)

Rock unit/ locality[a]	Comp.[b]/ colour	No. of samples (specimens)	Bedding corr.	Fold test[c]	NRM int.[d] (mA m^{-1})	Remanence direction				Pole					Palaeo-lat. (°)
						decl. (°)	incl. (°)	k	α_{95} (°)	lat. (°S)	long. (°E)	K	A_{95} (°)	Acr.	
Loc. 9+10	Red	21	No			339.3	−46.9	35.4	5.4	67.3	190.4	30.3	5.9		29.0
		(21)	Yes			*320.6*	*−50.6*	*29.3*	*6.9*	*50.9*	*195.3*	*21.0*	*8.3*		*32.3*
Loc. 9+10	Grey+	50	No			337.6	−52.2	17.7	4.9	63.2	180.3	12.5	5.9		34.8
	Red	(55)	Yes			*318.0*	*−55.1*	*17.8*	*4.9*	*47.4*	*188.8*	*12.1*	*6.1*	AME	*37.4*
Tatoola Sandstone					1–3 (L) 0.2–1 (U)										
Loc. 10	–	35	No			347.3	−50.0	10.7	7.8	69.5	170.8	7.6	9.4		33.4
		(35)	Yes			*326.5*	*−56.4*	*9.9*	*8.1*	*52.7*	*182.2*	*6.1*	*10.7*	TAT	*39.4*
Tooganinie Formation					0.5–4										
Loc. 10+11	–	55	No			354.0	−50.8	11.7	5.9	71.2	153.0	9.2	6.7		35.0
Neg. incl.		(60)	Yes			*330.9*	*−52.0*	*11.5*	*5.9*	*57.4*	*186.3*	*8.3*	*7.1*		*35.2*
Loc. 10+11	–	17	No			173.3	28.4	10.9	11.3	83.5	235.1	12.6	10.5		15.9
Pos. incl.		(19)	Yes			*163.5*	*43.9*	*11.9*	*10.8*	*72.0*	*188.3*	*10.7*	*11.5*		*27.4*
Loc. 10+11	–	72	No			353.8	−45.6	10.2	5.5	75.4	159.3	9.1	5.9		30.4
Combined		(79)	Yes	F+		*334.3*	*−50.1*	*11.5*	*5.2*	*61.0*	*186.7*	*8.5*	*6.1*	TOO	*33.4*
Myrtle Shale					0.3–7										
Loc. 12	I	26	No			351.7	−35.7	15.0	7.6	81.6	198.4	16.6	7.2		20.8
		(33)	Yes			*346.5*	*−38.6*	*13.1*	*8.2*	*75.9*	*197.4*	*14.6*	*7.7*	MYR	*23.4*
Emmerugga Dolomite															
Loc. 13	–	16	No		0.1–6 (L)	357.0	−43.2	7.5	14.5	78.6	162.8	6.1	16.3		27.1
		(16)	Yes		1–30 (U)	*342.4*	*−38.3*	*7.6*	*14.4*	*71.9*	*204.7*	*6.6*	*15.6*	EMMB	*22.7*
	O	106	No			*256.6*	*56.3*	*17.5*	*3.4*	*21.9*	*81.8*	*10.7*	*4.4*	EMMO	*39.1*
		(106)	Yes			246.2	71.1	18.2	3.3	28.0	102.8	8.3	5.1		57.6
Loc. 14	–	31	No		0.5–6	351.7	−16.7	23.9	5.4	78.8	270.4	31.1	4.7		8.8
		(31)	Yes			*352.1*	*−35.4*	*25.7*	*5.2*	*82.3*	*199.4*	*26.1*	*5.2*	EMMA	*20.1*
Loc. 13+14	–	47	No			353.1	−25.1	10.7	6.7	82.4	244.3	11.8	6.3		14.5
		(47)	Yes	F+		*349.1*	*−36.4*	*14.2*	*5.7*	*79.1*	*202.6*	*12.8*	*6.1*	EMM	*21.0*
Balbirini Dolomite					0.1–5										
Loc. 16	I	100	No			3.2	−38.5	18.8	3.4	83.3	108.3	16.8	3.6		22.7
		(103)	Yes	F+		*336.1*	*−55.6*	*19.8*	*3.3*	*60.6*	*176.1*	*11.7*	*4.3*	BDB	*37.6*
	O	38	No			*247.6*	*44.7*	*21.7*	*5.1*	*27.5*	*68.8*	*19.8*	*5.4*	BDBO	*27.8*
		(38)	Yes			279.6	78.8	21.8	5.1	12.2	115.8	8.0	8.8		70.2
Roper Group[f]					1–50										
Loc. 18+19	I	13	No			316.7	−5.1	9.7	14.0	−45.4	58.8	15.7	10.8		0.0
		(28)	Yes			*316.7*	*6.3*	*9.6*	*14.2*	*−42.9*	*66.7*	*15.3*	*11.0*	RGBI	*3.2*
	H	10	No			324.7	−4.1	8.6	17.5	−52.9	64.6	12.1	14.5		2.4
		(16)	Yes			*324.7*	*5.3*	*8.3*	*17.8*	*−50.5*	*71.3*	*12.0*	*14.6*	RGBH	*2.4*

Italics indicate alternative chosen because of field test results or as the preferred interpretation. k, K are precision parameters (Fisher, 1953); α_{93}, A_{95} are semi-angles of 95% cone of confidence.

[a] Locality numbers as in Figs. 1 and 2.

[b] I = intermediate unblocking temperature component; H = high unblocking temperature component; O = overprint.

[c] F+ denotes positive fold test; F− denotes negative fold test.

[d] General intensity range. U and L indicate intensity in the upper and lower part of formation/section.

[e] Partial bedding correction as discussed in text.

[f] Preliminary results.

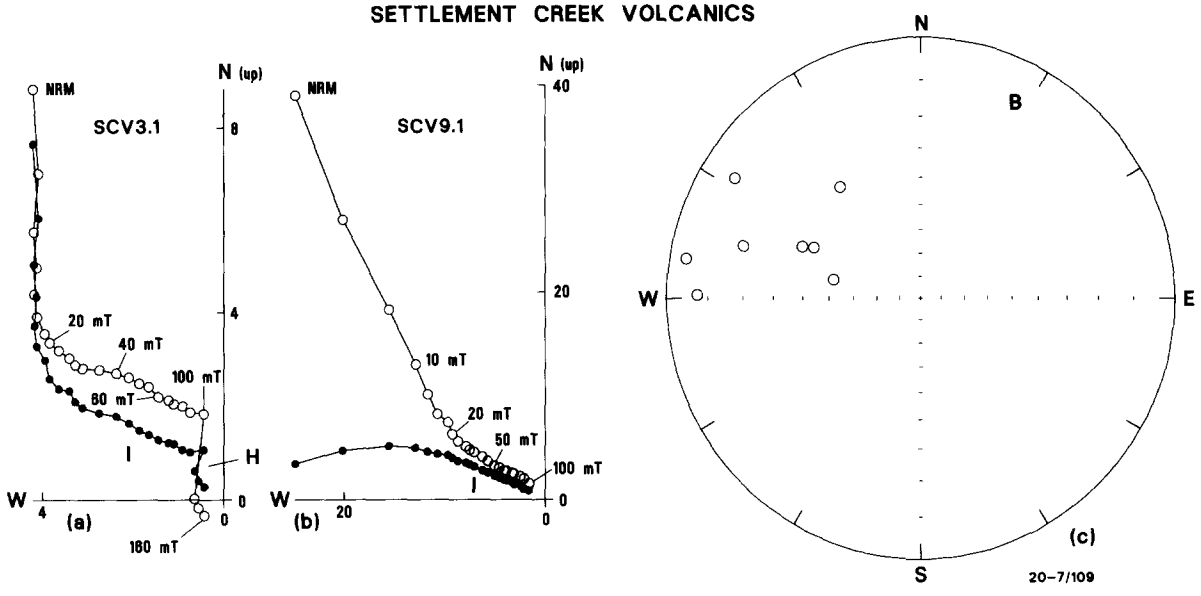


Fig. 4. Settlement Creek Volcanics. Examples of orthogonal projection plots for AF demagnetization showing remanences with (a) intermediate (I) and high (H) coercivity components, and (b) I component only. Plot (c) shows bedding-corrected directions for the I component. In the orthogonal projection plots open (filled) symbols indicate projections onto vertical (horizontal) plane; in the stereo plot open (filled) symbols indicate directions in upper (lower) hemispheres. B signifies bedding-corrected directions. Axial units in the orthogonal plots are mA m^{-1}.

bers of the Tawallah Group, crop out over a small area near the southeastern margin of the McArthur Basin. Twenty-six cores of pale- and medium-red rock were spaced roughly equally within a 30 m stratigraphic interval that crossed multiple flows at Locality 4. The bedding dip for the site was obtained from the immediately underlying Wollogorang Formation.

The NRM intensities of samples from the lower half of the sequence ranged from 5 to 50 mA m^{-1}, whereas those from the upper half ranged from 35 to 3000 mA m^{-1}. This difference was reflected in thermal demagnetization which gave grouped directions for the lower half, but highly scattered directions above (Fig. 7a). Since the upper sequence crops out on hill tops, lightning strikes seem a possible cause of the scatter. Therefore, the seven remaining specimens were demagnetised partly by AF before thermal treatment. This procedure removed a low-coercivity component (cf. Figs. 7c and 7d) and greatly reduced the scatter, leaving only one discordant direction (Fig. 7b).

Two positively inclined remanence components, I and H, were resolved in each "lower" sample; component I unblocked at ~580°C, and component H at ~680°C (Figs. 7e and 7f). Only H was resolved in the "upper" samples. The orthogonal plots of three samples contained evidence of a third component in the form of short linear segments between 580°C and 650°C (Comp. 3 in Fig. 7f). This component was not resolved, but it appears to have a northerly, negative inclination.

The combined bedding-corrected directions of the lower and upper parts of the sequence are plotted for components I and H in Figs. 7g and 7h), and give the respective means 103.1°, 37.1° ($\alpha_{95}=13.7°$) and 100.0°, 47.2° ($\alpha_{95}=11.1°$). The mean for H excludes the discrepant direction noted above. In specimens where both components were resolved, the I component was always shallower than H.

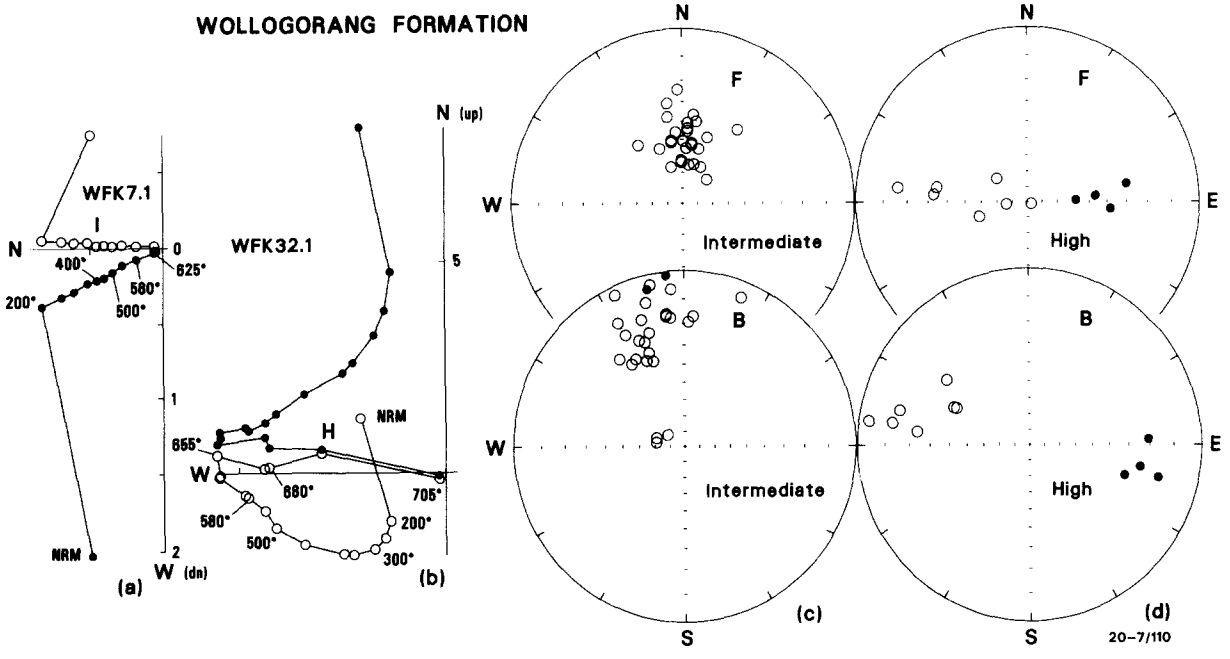


Fig. 5. Wollogorang Formation. Representative orthogonal projection diagrams showing (a) intermediate (*I*) and (b) high (*H*) unblocking temperature components. Stereo plots (c) and (d) show the I and H remanence directions before (*F*) and after (*B*) bedding corrections. Symbols as for Fig. 4.

Table 3
Fold tests

Rock unit	Pole acronym	No. of dirs.	Degr. of freedom	Test statistic				Conclusion
				before corr.	after corr.	95% crit. value	99% crit. value	
McElhinny (1964) fold test (test statistic: k_1/k_2)								
Wollogorang Fm.	WFK	11	20, 20		2.54	2.12	2.94	Positive: 95% level
McFadden and Jones (1981) fold test (test statistic: $[R_a+R_b-R^2/(R_a+R_b)]/2(N-R_a-R_b)$)								
Emmerugga Dol.	EMM	47	–	0.316	0.031	0.069	0.108	Positive: 99% level
McFadden (1990) fold test (test statistic: ξ_1 or ξ_2)								
Wollogorang Fm.	WFKO	27	–	4.72	10.98	6.04	8.55	Negative: 99% level[a]
Mallapunyah Fm.	MAL	122	–	53.85	2.50	12.85	18.17	Positive: 99% level[b]
Tooganinie Fm.	TOO	72	–	12.14	5.57	9.87	13.96	Positive: 95% level[a]
Balbirini Dol.	BALB	84	–	11.26	5.54	10.66	15.07	Positive: 95% level[a]

[a] Statistic ξ_1.
[b] Statistic ξ_2.

3.4. Packsaddle Microgranite

This medium-red microgranite was sampled at Localities 5 and 6. For Locality 5, the bedding dip was assumed to be the same as that for the Hobblechain Rhyolite, ~3 km to the south-

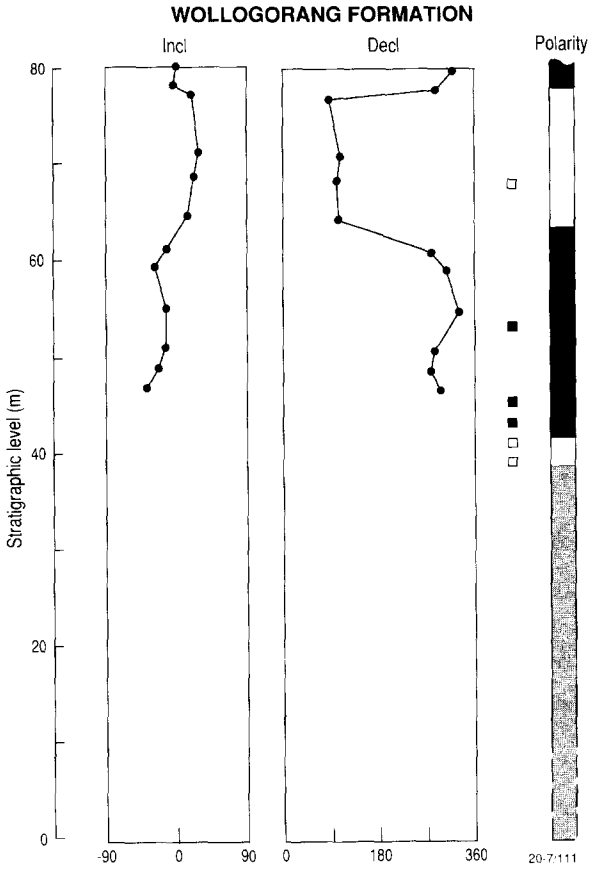


Fig. 6. Polarity reversal column for the Wollogorang Formation. In this and subsequent reversal columns, the polarity associated with easterly and southeasterly directions is shown white, its reverse is shown black, and parts of column with undetermined polarity are shown grey. Square symbols next to the reversal column indicate polarity data from samples where complete separation of primary remanence from overprints was not possible due to overlapping blocking temperature spectra; so primary directions are not available. Dotted line at the base of the column indicates missing section, estimated as about 10 m thick.

southwest. For Locality 6, bedding dips were obtained from nearby outcrops of the Wollogorang Formation.

A total of nineteen samples was collected. At Locality 5, thermal demagnetization directions were highly scattered; at Locality 6 they grouped except for two samples with well above average NRM intensities. AF, then thermal, demagnetization of samples from Locality 5 yielded a high unblocking temperature component with concordant directions for all but one sample.

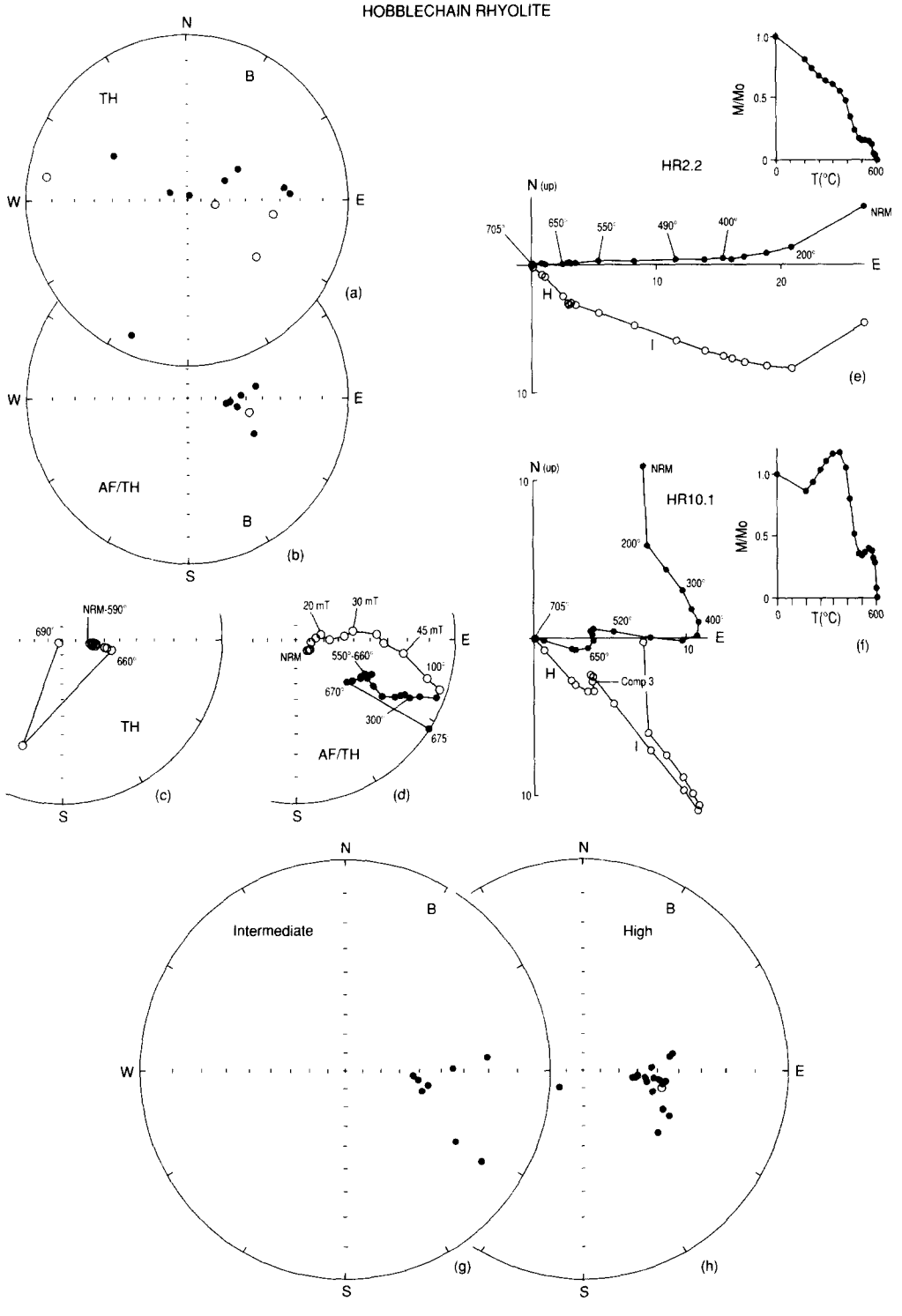
As for the Hobblechain Rhyolite, the measurements yielded components I and H of unblocking temperatures $\sim 580°C$ and $\sim 680°C$ (Figs. 8a and 8b). The I directions, resolved in only six pilots, were highly scattered (Fig. 8c), so their mean is considered unreliable. Fig. 8d shows the H directions. Excluding the single discordant direction, the bedding-corrected mean is 96.1°, 50.8° ($\alpha_{95}=8.5°$), in good agreement with the mean H direction for the Hobblechain Rhyolite.

3.5. Masterton Sandstone

The Masterton Sandstone, lowermost formation of the McArthur Group, was sampled at Locality 7. The sequence comprises predominantly red-brown quartz sandstones that grade from fine- or medium-grained in the lower part of the sequence to coarse-grained at the top. A 5 m zone immediately below the overlying Mallapunyah Formation is heavily ferruginized and silicified, and its characteristic dark, red-brown mottles extend 15 m below the contact. Seventy-one samples were collected over a stratigraphic interval of 50 m.

The remanence directions from the top 16 m

Fig. 7. Hobblechain Rhyolite. Stereo plots at top left show remanence directions in the upper half of this formation obtained by (a) thermal demagnetization and (b) partial AF demagnetization followed by thermal demagnetization. The contrasting responses of specimens from the same sample to the two demagnetization procedures are illustrated in (c) for thermal and (d) for AF/thermal demagnetization. The orthogonal projection and normalised intensity plots show examples of thermal demagnetization for (e) a specimen with intermediate (I) and high (H) unblocking temperature components and (f) a specimen that contains a third, unresolved component. Both samples are from the lower part of the sequence. Stereo plots at the bottom show remanence directions from all parts of the sequence for (g) intermediate and (h) high unblocking temperature components. Symbols as in Fig. 4.


HOBBLECHAIN RHYOLITE
N
B
TH
W
E
(a)
W
E
AF/TH
B
S
(b)
NRM-590°
690°
660°
TH
S
(c)
20 mT
30 mT
45 mT
NRM
550°-660°
100°
670°
300°
675°
AF/TH
E
S
(d)
HR2.2
M/Mo
T(°C)
N (up)
705°
650°
550°
490°
400°
NRM
200°
E
H
I
(e)
HR10.1
N (up)
NRM
200°
300°
400°
705°
520°
650°
E
H
Comp 3
I
(f)
Intermediate
High
(g)
(h)

of the formation, including the heavily ferruginized zone, differed from the rest and are described separately below. Generally, the other directions were poorly resolved because of remanence instability at higher temperatures. Only eighteen samples had well-defined directions, giving a component H that unblocked at ~680°C. In seventeen of the samples H had shallow inclinations to the northwest; in the eighteenth, H was inclined to the southeast (Fig. 9d). The bedding-corrected mean, after inversion of the SE direction, is 288.2°, −5.3° (α_{95}=8.1°). This is similar to the mean directions of the upper Tawallah Group (Table 2).

Several samples contained a second component I (Figs. 9a and 9b) with a direction similar to H, but an unblocking temperature of ~580°C. Components I and H are probably coeval and, from their unblocking temperatures, appear to have magnetite and hematite as carriers. The four samples with well-defined I direction are too few for a reliable mean.

Samples from the ferruginous top 16 m of the sandstone had a single-component remanence of hematite-range unblocking temperature. The directions were well-defined (Fig. 9c) and well-grouped (Fig. 9e), with a mean 5.4°, −31.1° (α_{95}=8.2°) that differs widely from the mean directions in adjacent sequences. The directional disagreement and Fe mottling suggest that the component is a chemical overprint.

Although the directions of many samples were poorly defined, most could be assigned polarities. The reversal column for the Masterton Sandstone is shown in Fig. 10.

3.6. Mallapunyah Formation

The stratigraphic thickness of the Mallapunyah Formation is about 220 m at Locality 8 where most samples were collected. The rocks are mainly red and purple dolomitic siltstones and fine- to coarse-grained sandstones. Desiccation cracks, intraclast breccias and occasional faint mottling suggest peritidal deposition and the possibility of hiatuses. Early diagenetic activity is suggested by the presence locally of small nodules of dolomite/barite and nodules, up to 1 m high, of "cauliflower chert" after anhydrite that have deformed the enclosing beds (Muir, 1979). The formation has been interpreted as primarily a supratidal continental sediment deposited in a subsiding region (Jackson et al., 1987).

A total of 358 samples was collected at Locality 8. In order to confirm the reversal pattern obtained from the upper part of the formation and to apply a fold test, 62 additional samples were collected from a duplicate section at Locality 9, about 3.5 km to the northeast.

The remanence directions at Localities 8 and 9 are shown in Figs. 11d and 11e. Except in a few samples with NRM intensities less than 0.5 mA m^{-1}, the directions in the lower 100 m of the sequence at Locality 8 were usually well-defined, with moderate positive inclinations to the southeast (Fig. 11a). The SE component was accompanied at the base, and occasionally higher, by a component of lower unblocking temperature and northerly negative inclination (Fig. 11b). From 100 to 125 m, reliable directions were not usually obtained; however, the presence of the SE component was evident up to the 160 m level. Above 160 m the remanences had northerly negative inclinations, except in a narrow interval containing the SE component (Fig. 11c).

The SE directions at Locality 8 form a scattered, but symmetrical group (Fig. 11d), whereas the directional grouping in the NW quadrant is elongate. Moreover, the inverted mean of the SE directions (Fig. 11d, cross symbol) falls outside the NW group, with the latter extending towards the mean. These features suggest the presence of two components in the NW group, one being the reverse of the SE components.

The elongation is largely due to 15 samples from the top 25 m of the formation (Fig. 11d, triangular symbols), representing a lithological change from carbonates (below) to interbedded sandstones and mudstones. With one exception, the fifteen directions form a close NNW cluster furthest from the inverted mean. Also, they resemble the directions of an overprint of same polarity at the base of the Wollogorang Formation (Fig. 5c). The topmost remanence is therefore interpreted as an overprint.

As discussed below, a NNW overprint is found

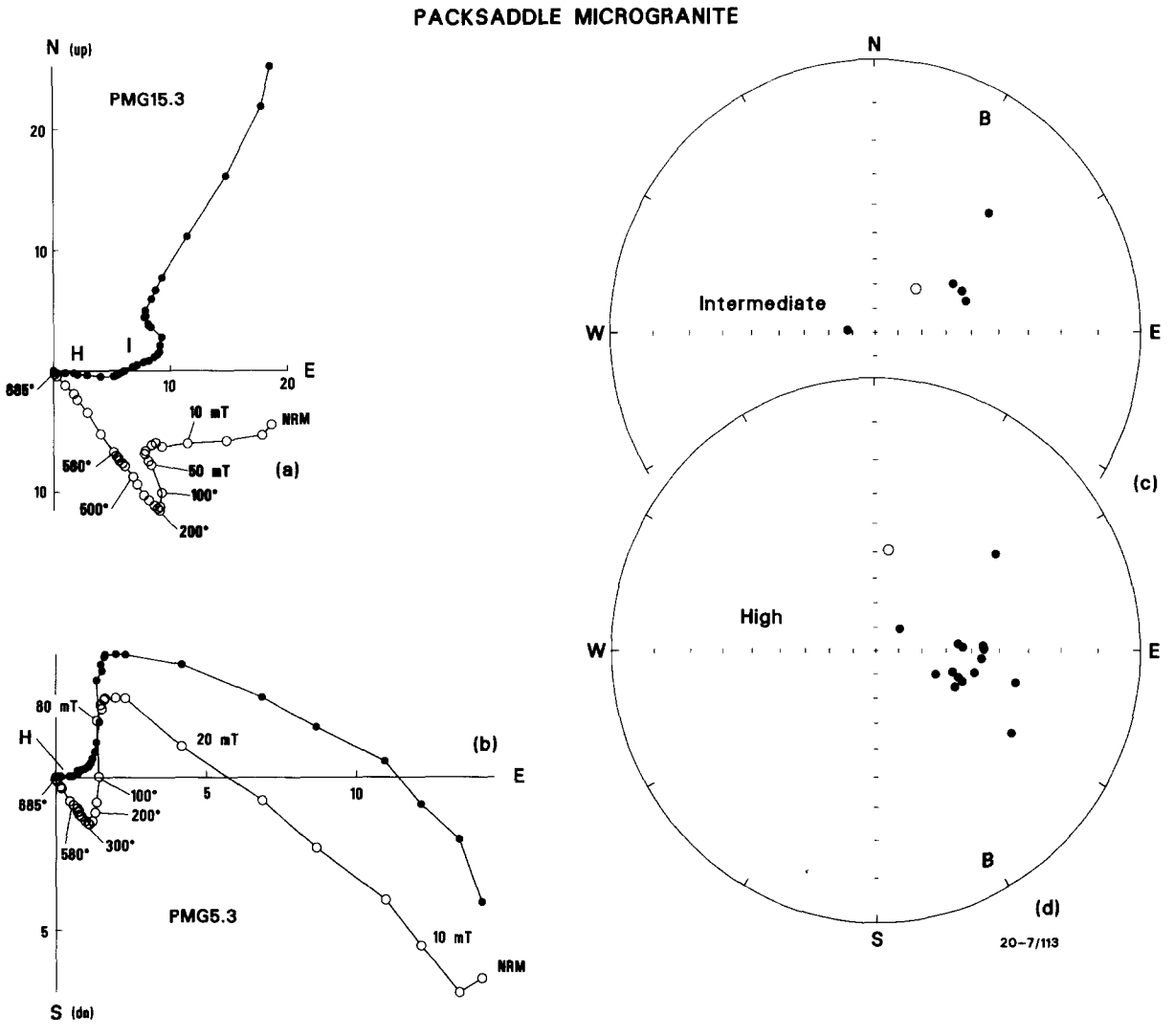


Fig. 8. Packsaddle Microgranite. Examples of orthogonal projection plots (a) and (b) showing intermediate (*I*) and high (*H*) unblocking temperature components. Directions from low-coercivity/low-temperature components (not annotated) are highly scattered. Stereo plots show remanence directions of (c) intermediate and (d) high unblocking temperature components. Symbols as in Fig. 4.

in the adjacent Amelia Dolomite where it occurs, as in the Wollogorang Formation, next to a prominent fault. This suggests that the remagnetization is related to fault zones (see Discussion). The basal Amelia Dolomite and the upper 25 m of the Mallapunyah Formation are completely remagnetized. Partial overprinting appears to extend down-sequence into the dolomitic part of the Mallapunyah Formation: the remanences there are directed mostly to the north-northwest, and the removal of the top-fifteen directions reduces but does not eliminate the asymmetry between the SE and NW quadrants. This is confirmed by a negative McFadden and McElhinny (1990) reversal test at the 99% level. As noted earlier, the effects of the NNW overprint in the Wollogorang Formation also extend beyond the zone of complete remagnetization. Because of the likelihood of bias, the NW quadrant directions at Locality 8 were not used for pole calculation.

The fold test of McFadden (1990) was ap-

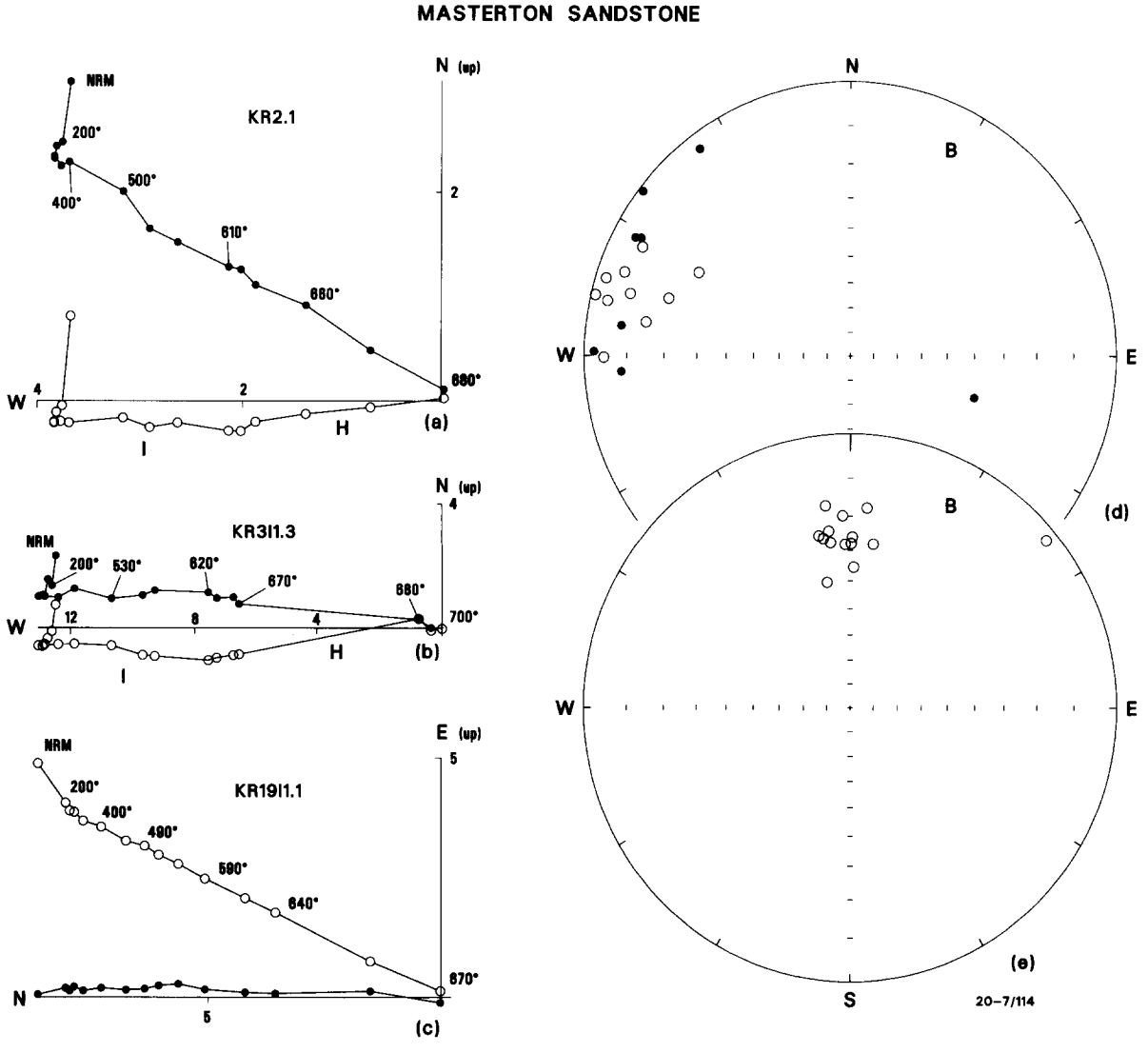


Fig. 9. Masterton Sandstone. Plots (a) and (b) show examples of orthogonal projections for the main part of the sequence which contains two components I and H of closely similar directions. Plot (c) shows a representative example for the ferruginised top of the sequence where only one remanence component was obtained. Stereo plots show remanence directions for (d) high unblocking temperature component (H) in the main sequence and (e) single-component remanence at the top of the sequence. Symbols as in Fig. 4.

plied to Localities 8 and 9, using bedding variations both within and between the localities. The test shows a strong correlation before bedding correction and none thereafter (Table 3). This suggests that the remanence predates the deformation which occurred presumably during block faulting of unknown age that produced the present-day horst. Fold tests were attempted also on sediments deformed plastically during the growth of large botryoidal quartz nodules: two basin-shaped dololutite layers were sampled, but at both the remanence directions were ill defined. The bedding-corrected mean directions for the SE and NNW groups are 122.0°, 33.0° ($\alpha_{95}=3.1°$) and 351.3°, −38.2° ($\alpha_{95}=8.9°$).

In Fig. 12, the reversal columns for Localities

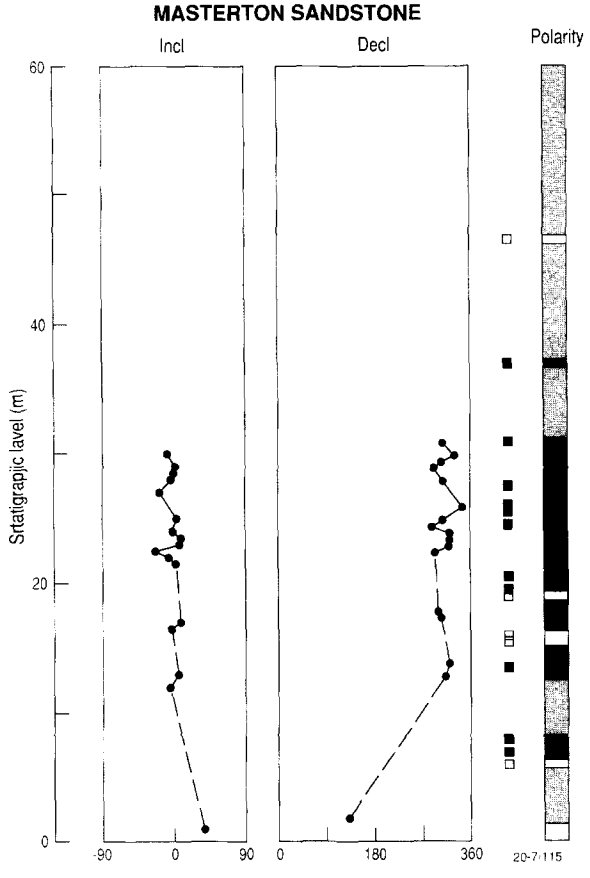


Fig. 10. Reversal column for Masterton Sandstone at its type section. Notation as in Fig. 6.

8 and 9 are similar, except for an extra reversal near the 100 m level at Locality 9 that is based on only one sample. The reversals of Locality 9 fill a gap in the record of Locality 8 where stream erosion and weathering have removed about 25 m of section. The upper sequences at both localities are overprinted too strongly for polarity estimation; otherwise the reversal record for the Mallapunyah Formation is complete for this region.

3.7. Amelia Dolomite

The Amelia Dolomite consists of alternate intervals of massive stromatolitic dolostone with interbedded dololutites, and dolarenite. The unit is interpreted as a peritidal deposit of either a marginal marine or continental sabkha environment.

The Dolomite was sampled principally at a 170 m composite section (Locality 9) from which 192, mostly grey or red-brown, samples were collected. Another 41 samples were collected from the basal 35 m of red-brown Amelia Dolomite at Locality 8, about 3.5 km to the southwest.

At Locality 9, most grey samples with NRM intensities under 0.5 mA m^{-1} became magnetically unstable during heatings, failing to give reliable remanence directions although their polarities were usually discernible. The magnetically stable grey samples unblocked at 500–580°C (Fig. 13a) and yielded single-component remanences with negative inclinations to the northwest. For the red-brown samples the demagnetization behaviour depended on the part of the sequence they were obtained from. Below the 140 m stratigraphic level they generally demagnetized at ~680°C and, with very few exceptions, yielded a single remanence component of steep positive inclination to the west-southwest (Fig. 13b). This is the first appearance in the succession of an overprint that recurs in the overlying McArthur and Nathan Group sequences. Above the 140 m level, the red-brown samples yielded a NW negative-inclination component similar to that in the grey samples. The remanences of the topmost red-brown samples differed from the others by an increasingly unstable magnetization above 500°C (Fig. 13c).

In contrast to the positive inclinations at the base of Locality 9, the basal Amelia Dolomite at Locality 8 contained a component with a northerly negative inclination. This resembles closely the NNW component in the immediately underlying upper Mallapunyah Formation, and is also interpreted as an overprint. Altogether three groups of remanence directions, including two overprints, are therefore identified in the Amelia Dolomite.

Because of extensive overprinting, the reversal column for the formation is incomplete (Fig. 14). Nevertheless, with the identification of a single narrow reversed interval, the formation appears to have mainly one polarity.

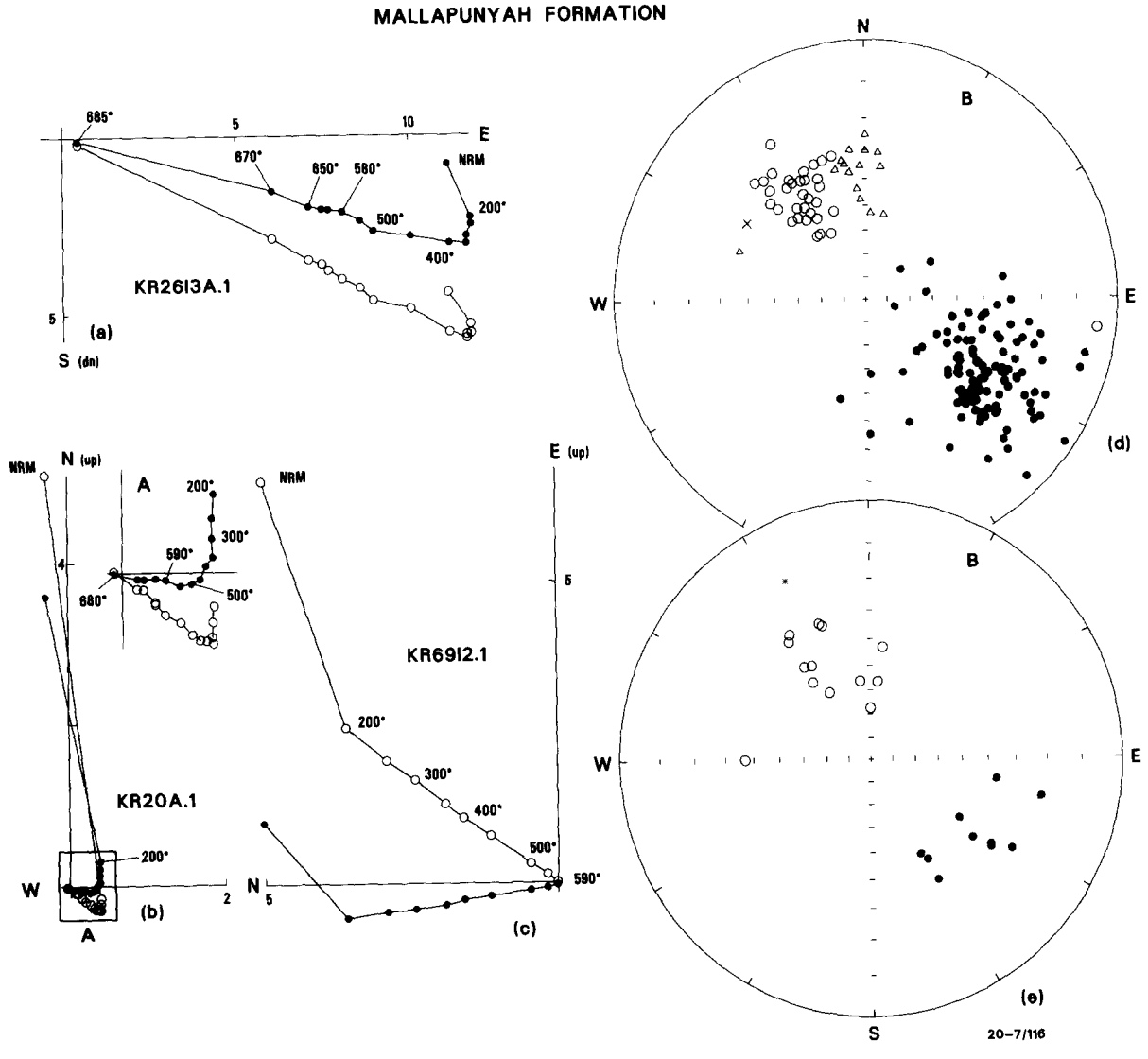


Fig. 11. Mallapunyah Formation. Orthogonal projection plots show examples of (a) component with moderate positive inclination to the southeast found in the lower part of the formation, (b) the SE component and a northerly negative-inclination component, and (c) a northerly negative-inclination component found high in the sequence. Stereo plots show remanence directions at (d) Locality 8 and (e) Locality 9. Triangular symbols in (d) are for directions from the top of the sequence, considered to represent an overprint; cross indicates the mean direction, after inversion, of the positive-inclination group. Note streaking of directions towards cross. Symbols as in Fig. 4.

3.8. Tatoola Sandstone

The lower two-thirds of this formation is composed of mainly fine-grained quartz sandstones, and the upper third of medium- to coarse-grained dolomitic sandstones with occasional stromatolites. The formation is interpreted as a marine sediment whose deposition was initiated by a rapid transgression, followed by gradual return to shallow-water environment. The transgres-

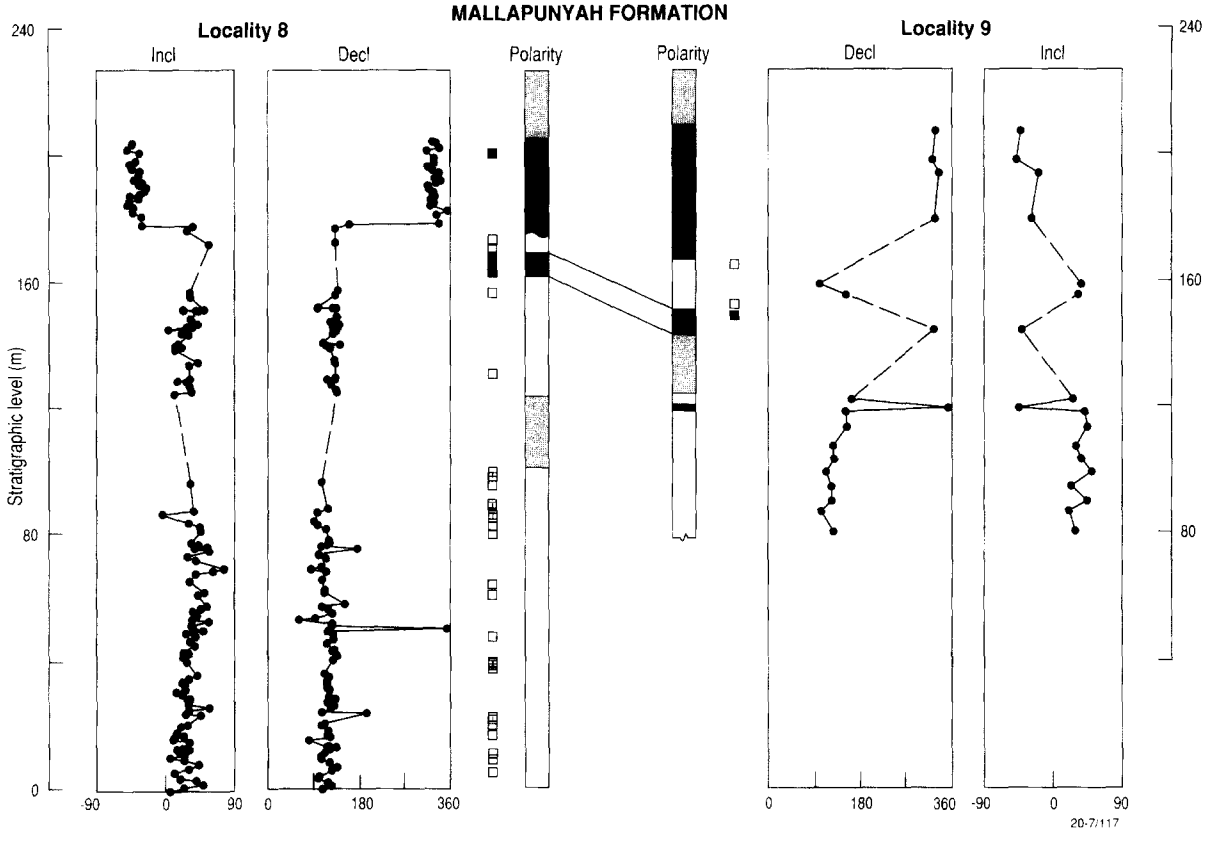


Fig. 12. Reversal column for Mallapunyah Formation at Locality 8 (type section) and Locality 9. Notation as in Fig. 6.

sion appears to have been due to uplift of adjacent sandstones (e.g., Tawallah Group or the oldest formations of the McArthur Group) that provided the clastics for the basal sequence (Jackson et al., 1987).

A total of 72 samples was collected from a 125 m thick sequence at Locality 10. NRM intensities were typically 1–3 mA m^{-1} in the mostly pale red-brown quartzose lower two-thirds, and 0.2–1 mA m^{-1} in the grey dolomitic upper third of the sequence. The unblocking temperatures were generally $\sim 600°C$, though some red-brown samples unblocked at $\sim 680°C$. Parts of the sequence had unstable remanences or gave poorly defined directions. The well-defined directions (e.g., Figs. 15a and 15b) plot as a loose group (Fig. 15c), giving a component with the bedding-corrected mean 326.5°, −56.4° (α_{95} = 8.1°).

Unstable remanences and discontinuous outcrop cause many gaps in the polarity column for the Tatoola Sandstone (Fig. 16). Except for one interval, no reversals are evident.

3.9. Tooganinie Formation

The Tooganinie Formation, like the Amelia Dolomite, consists of interbedded dolostones, shales and dololutites. These formations contain stromatolites and have similar sedimentary structures, suggesting the return, after deposition of the Tatoola Sandstone, to the Amelia Dolomite depositional environment.

The formation was sampled at Localities 10 and 11 where the top of the 240 m thick sequence is slightly truncated by faulting. Bedding dips change from northerly at the base to easterly near the top, providing suitable conditions for a fold test. The sequence is predominantly grey, with occasional olive-green and red-brown parts.

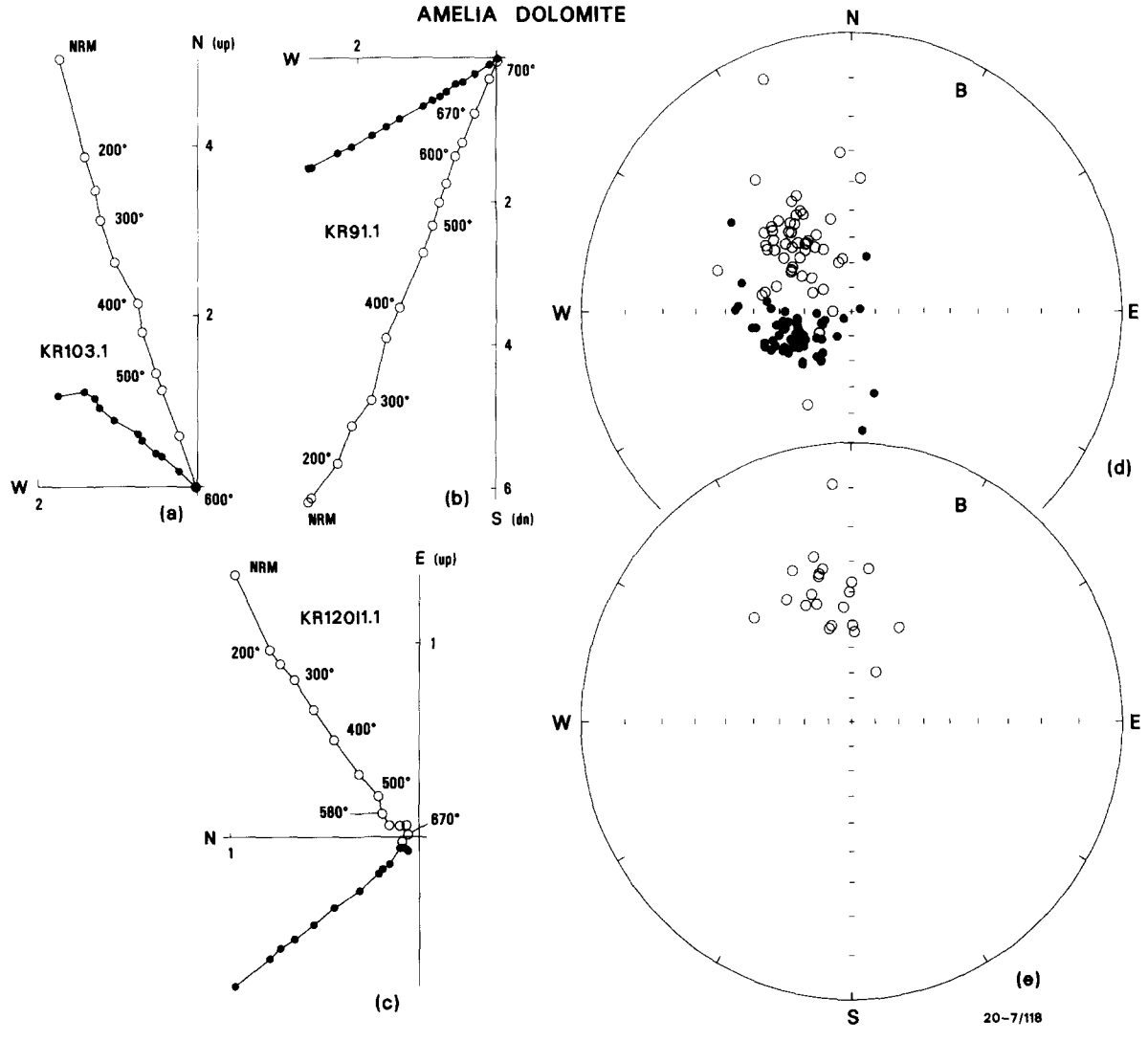


Fig. 13. Amelia Dolomite. Orthogonal projection plots show examples of single-component remanences in (a) grey samples, (b) red-brown samples from the lower part of the sequence, and (c) red-brown samples from the top part. Stereo plots show the remanence directions at (d) Locality 9 and (e) Locality 8. Symbols as in Fig. 4.

A total of 163 samples was collected. Unblocking temperatures were generally ~580°C, but a few exceeded 600°C. Many samples yielded poorly defined remanence directions. Those with well-defined directions (Figs. 17a and 17b) had two components, characterised by NNW negative and SSE positive inclinations (Fig. 17c). The mean directions are statistically indistinguishable, so the components are assumed to represent opposite polarities of the same geomagnetic field. McFadden's (1990) fold test was positive at the 95% confidence level (Table 3): therefore, the remanence appears to be pre-folding. The combined bedding-corrected mean direction was 334.3°, −50.1° ($\alpha_{95}=5.2°$).

The reversal column for the Tooganinie Formation contains gaps due to unstable remanence and lack of outcrop (Fig. 18). In contrast to the underlying formations, this formation contains many reversals.

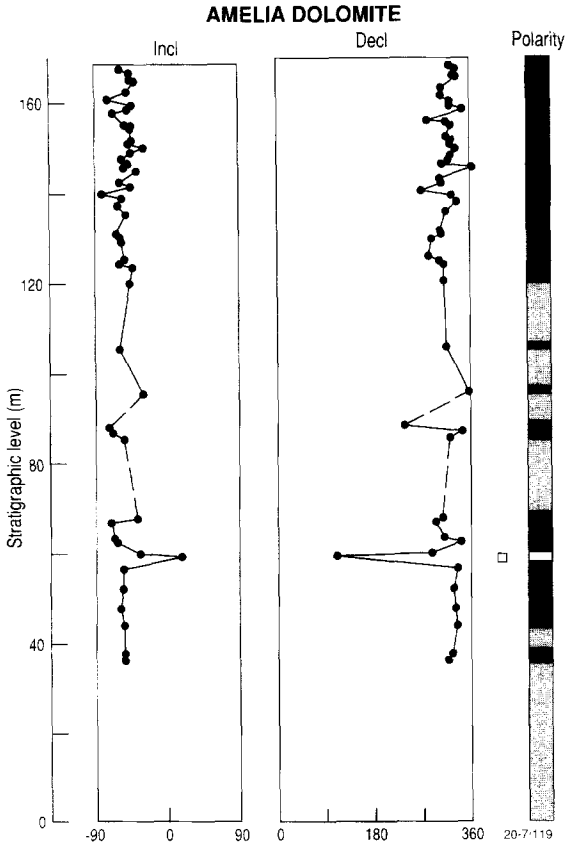


Fig. 14. Reversal column for Amelia Dolomite at Locality 9 (type section). Notation as in Fig. 6.

3.10. Myrtle Shale

The Myrtle Shale consists predominantly of red-brown siltstones and fine-grained quartz sandstones, with halite casts throughout. It resembles lithologically the lower part of the Mallapunyah Formation and is likewise interpreted as a lacustrine or low-energy fluvial deposit, with a possible aeolian contribution to the finer grain-size fraction.

Forty-one samples were collected from the 60 m thick type section at Locality 12 where gaps in exposure amount to ~20% of the thickness. Most samples were either dark red-brown or pale-grey; a few were pale-brown or pale red-brown. Several grey samples contained red laminations.

The remanence directions were generally well-defined up to demagnetizing temperatures of 600°C, but became increasingly scattered thereafter. The scatter was accompanied by a steep increase in magnetic susceptibility, suggesting the production of a new magnetic phase during heating.

The grey specimens contained a single-component remanence, I, that unblocked at ~600°C and had a negative northerly inclination (Fig. 19a). Most other pale-coloured specimens and a few of the dark red-brown specimens yielded directions similar to I. The remaining specimens, including all with red laminations, contained a second component, H, of higher unblocking temperature. H was rarely resolved, but from demagnetization trends it appears to have a positive and moderately steep inclination (Fig. 19b). Like directionally similar components in the immediately overlying Emmerugga Dolomite and in parts of the Balbirini and Amelia Dolomites, H is interpreted as an overprint.

The stereo plot for the I component is scattered (Fig. 19c), possibly partly because of incomplete removal of component H. Excluding four discrepant directions, its bedding-corrected mean is 346.5°, −38.6° ($\alpha_{95}=8.2°$). No reversals were found in the I component (Fig. 20).

3.11. Emmerugga Dolomite

This stromatolitic formation comprises mainly interbedded dololutites and dolostones, and has a maximum thickness of 620 m. The lower Mara Dolomite Member, which forms the bulk of the formation, contains abundant sedimentary structures. The upper Mitchell Yard Dolomite Member is mostly massive due to extensive vadose alteration, which occurred apparently before the deposition of the overlying Teena Dolomite (Muir, 1983). The Emmerugga Dolomite is silicified in many higher parts of the landscape, and its iron concentration is well above average near the HYC deposit (Brown et al., 1969). The formation has been interpreted as either a shallow saline lake or marginal marine deposit.

The Emmerugga Dolomite was sampled at its type section (Locality 14) and at Locality 13, 33 km to the southeast. At the type section the rocks are red-brown at the base and predominantly grey

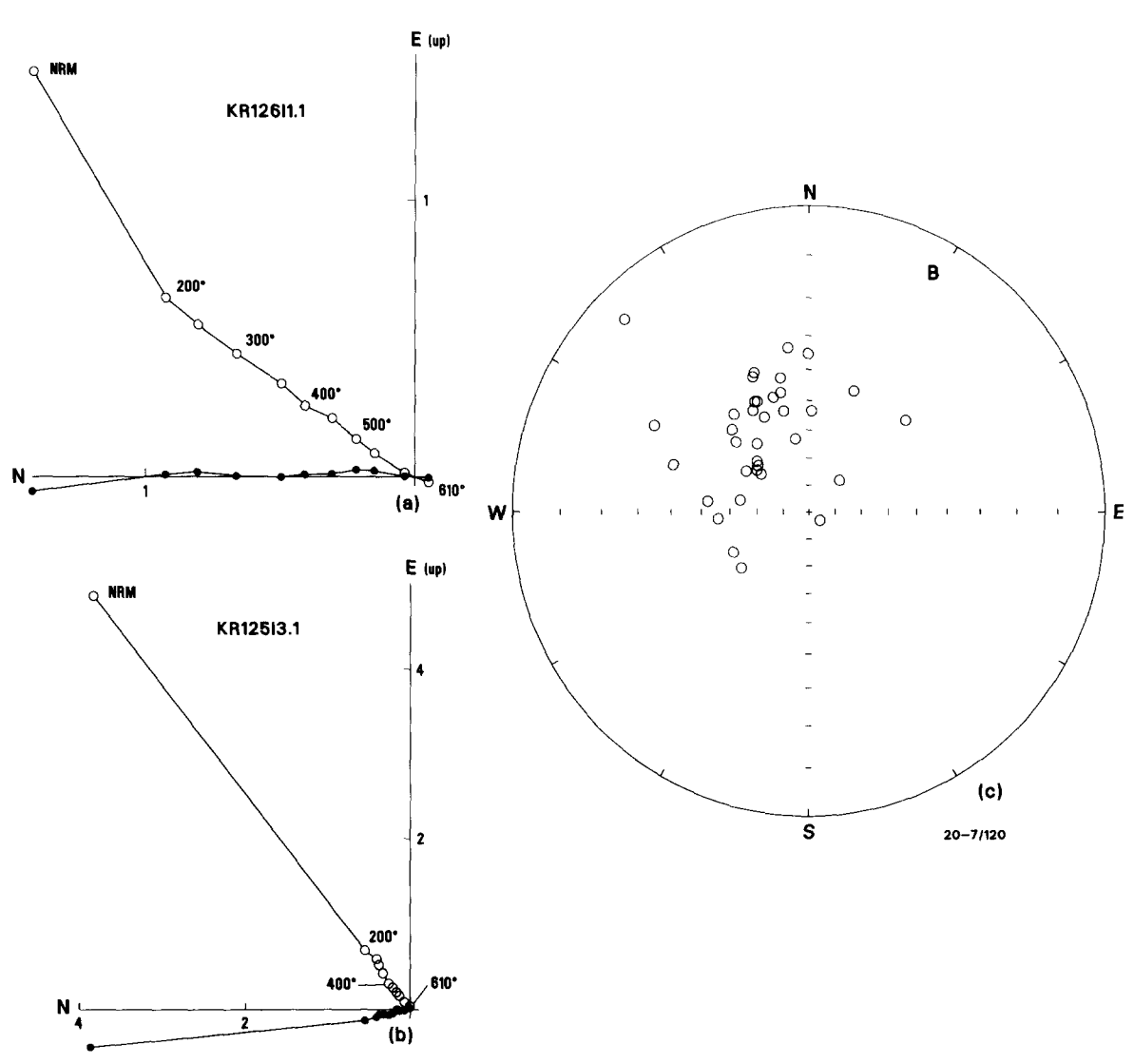


Fig. 15. Tatoola Sandstone. (a) and (b) show representative examples of orthogonal projection plots; (c) shows the bedding-corrected remanence directions. Symbols as in Fig. 4.

elsewhere; at Locality 13, where only the lower two-thirds of the Mara Dolomite Member crops out (Jackson et al., 1978), the formation is red- to purple-brown, with pale red-grey bedding intervals in the lower part.

At the type section, the 90 samples that were collected have NRM intensities ranging from 0.5 to 6 mA m^{-1}, except at the red base where the intensities reach 140 mA m^{-1}. Thermal demagnetization used alone yielded orthogonal projection plots with well-defined linear segments but random directions (Fig. 21e). However, when thermal demagnetization was preceded by partial AF demagnetization, three distinct components were resolved (Fig. 21b): a low-coercivity component L1 causing the randomness; component L2, probably due to goethite, unblocks by 100°C and has a direction (1.6°, −48.4°) similar to the present geomagnetic field direction (4.8°, −47.8°, Figs. 21f and 21g); and high-

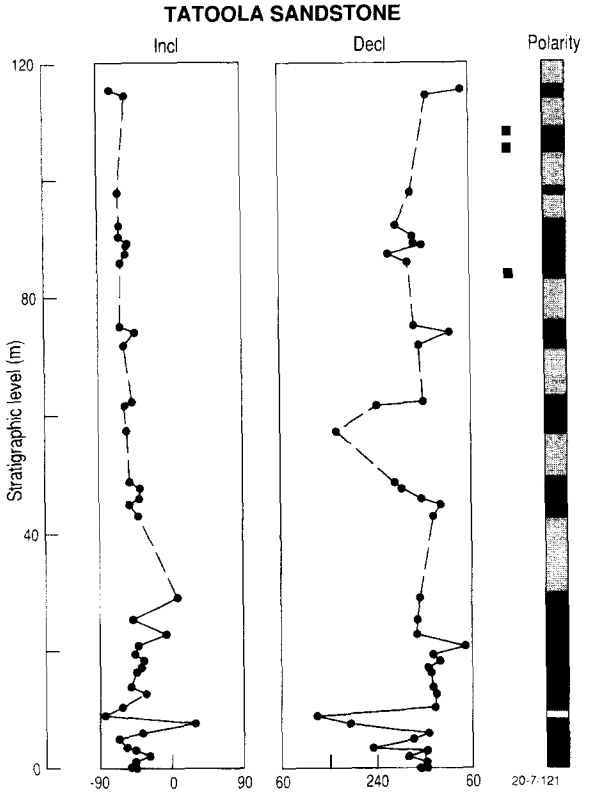


Fig. 16. Reversal column for Tatoola Sandstone at Locality 10 (type section). Notation as in Fig. 6.

temperature component H. Only H (Fig. 21h) with the bedding-corrected mean direction 352.1°, −35.4° ($\alpha_{95}=5.2°$) is geologically significant. H was poorly defined near the top of the Mara Dolomite Member and in the overlying recrystallised Mitchell Yard Dolomite Member. Therefore, the acceptance criterion for the latter unit was relaxed from MAD < 5° to MAD < 10°, giving the eight directions in Fig. 21i. Their similarity, despite recrystallization, with the other H directions (cf. Figs. 21i and 21h) is consistent with alteration of the Mitchell Yard Dolomite before the overlying Teena Dolomite was deposited (Muir, 1983): i.e., the primary and secondary component ages are similar.

At Locality 13, the 196 samples gave two remanence components (Fig. 21j). The first (Fig. 21c), found only in the lower parts of the sequence, had highly scattered directions but was otherwise similar to the H directions at the type section and the I directions in the underlying Myrtle Shale. Its unblocking temperatures were mainly 500–580°C, though some reached 680°C. This component gave the bedding-corrected mean direction 342.4°, −38.3°, ($\alpha_{95}=14.4°$). The second, predominant, component unblocked at ~680°C and had generally very well-defined directions with moderate to steep positive inclinations to the west-southwest (Fig. 21d). These plot as a tight group (Fig. 21j) and give the field-corrected mean 256.3°, 56.3°, ($\alpha_{95}=3.4°$).

The beds at Localities 13 and 14 have an interdip angle of about 30°. McFadden and Jones' (1981) fold test, applied to the H component at Locality 14 and its counterpart at Locality 13 (Figs. 21k and 21l), showed with 99% confidence that the remanence is pre-folding (Table 3). Its bedding-corrected mean direction is 349.1°, −36.4° ($\alpha_{95}=5.7°$).

The second component at Locality 13 is directionally closely similar to overprints in the Amelia Dolomite, Myrtle Shale and Balbirini Dolomite (see below). As its pole plots out of sequence on the path, this component is interpreted as an overprint. It has obliterated most primary remanence at Locality 13.

The reversal columns for the type section and the lower, least overprinted, parts of Locality 13 contain only one reversed interval each, at the base of the column (Fig. 22).

3.12. Balbirini Dolomite

The Balbirini Dolomite, the older of the two Nathan Group formations, has a maximum thickness of ~1500 m. Its basal beds are conglomerates, sandstones and siltstones. These are overlain by sandstones, siltstones and dolostones, which are initially evaporitic and become progressively stromatolitic higher up. The upper parts of the formation have been largely recrystallised, apparently before deposition of the Roper Group. The basal conglomerate is thought to be a tectonic talus deposit, and the main evaporitic and stromatolitic sequence is interpreted as a sabkha deposit.

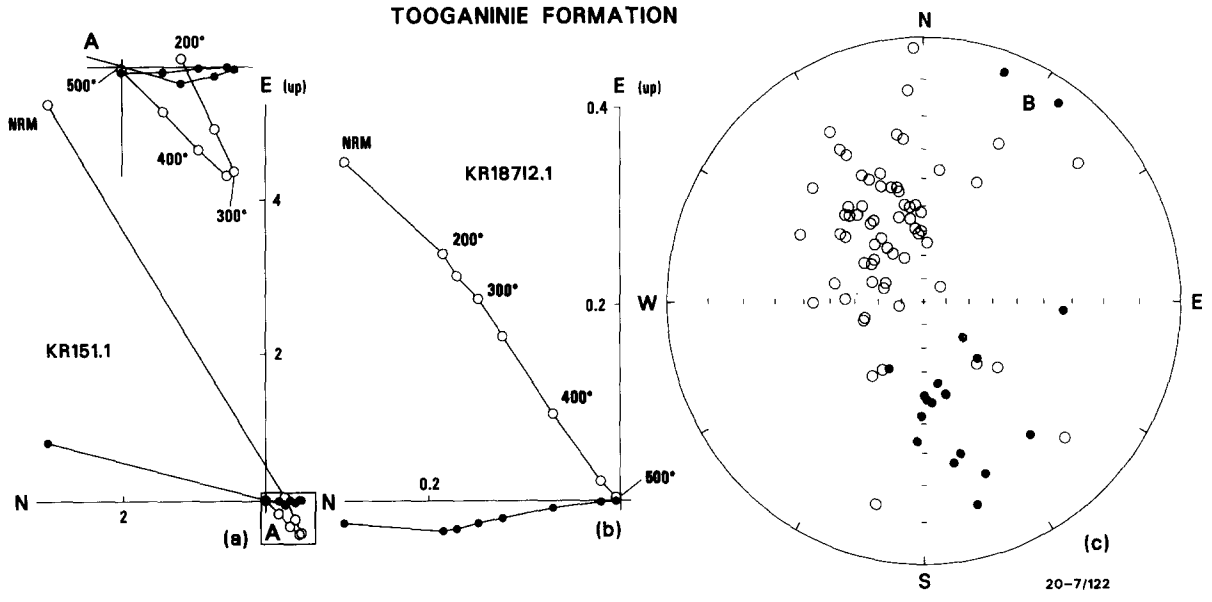


Fig. 17. Tooganinie Formation. Orthogonal projection plots showing examples of (a) positive- and (b) negative-inclination components. Stereo plot shows bedding-corrected remanence directions. Although several directions are discrepant ("intermediate"), negative- and positive-inclination groups are evident. Symbols as in Fig. 4.

The Balbirini Dolomite was sampled at its 2.5 km long type section (Locality 16), which lacks the top 150 m interval known from drilling. The section is semi-continuous with narrow resistant bands separated by recessive units, and spans ~900 m of stratigraphic thickness. The 257 samples are mostly grey, except those from the basal 70 m and top 250 m which are predominantly red- or purple-brown. Block rotation and irregular bedding occasionally prevented accurate dip estimation. However, bedding dips generally range up to 15–20°, and indicate open folding about east–west axes.

Two remanence components were obtained. The first, from the unrecrystallised lower part of the sequence, is bipolar with moderate negative inclinations to the north-northwest or positive inclinations to the south-southeast (Figs. 23a and 23d). Its unblocking temperature was typically ~580°C for the grey samples and ~680°C for the red samples. After exclusion of directions associated with highly anomalous bedding dips, McFadden's (1990) fold test was positive at the 95% confidence level (retention of the anomalous data reduced this level to just below 95%). The bedding-corrected mean direction is 336.1°, −55.6° ($\alpha_{95}=3.3°$).

The second component (Fig. 23c) was found only in the recrystallised upper part where it is characterised by a moderately steep positive inclination to the west-southwest (Fig. 23e). McFadden's fold test was inconclusive. Its mean direction before bedding correction is 247.6°, 44.7° ($\alpha_{95}=5.1°$).

The reversal column for the lower two-thirds of the formation is based on sparse data, but appears to comprise predominantly one polarity with numerous brief reversed intervals (Fig. 24). Because of recrystallization, few polarity data are available for the upper third of this unit.

3.13. Roper Group

The Roper Group, which appears from isotopic evidence and lithological contrast to have been deposited long after the carbonate sequences, was sampled in reconnaissance style. All 32 samples were from its lower formations: the Limmen Sandstone and Crawford Formation. The Limmen Sandstone was sampled at Locality

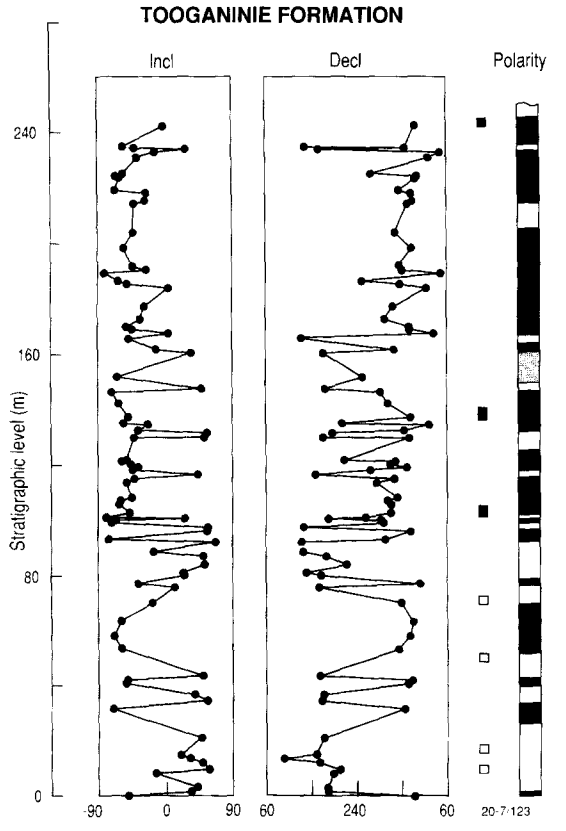


Fig. 18. Reversal column for Tooganinie Formation at Localities 10 and 11 (type section). Notation as in Fig. 6.

18 where it is pale-brown and highly silicified, and the Crawford Formation was sampled at Locality 19, a cliff face. At Locality 19, the upper part of the Crawford Formation contains fractures with yellow to red aureoles and fine, purple-red mottles, suggesting remobilization of iron. Although similar colouring was observed along some bedding planes at the base of the cliff, the rock there was predominantly pale-brown.

The remanence directions were highly scattered (Fig. 25a), most of the scatter coming from the red samples. With few exceptions, the directions from the other samples were inclined at shallow to moderate angles towards either the northwest or southeast. In most samples two remanence components of unblocking temperatures ~580°C and ~680°C were resolved (Figs. 25c and 25d), suggesting magnetite and hematite as the respective remanence carriers. The directional difference of the components is small but distinct, an inflection in the intensity–temperature demagnetization plots usually marking the removal of the lower-temperature component (Fig. 25b). The cause of the small directional difference is not clear, but it could be, for example, systematic errors of one or both of the minerals in recording the same palaeogeomagnetic field, or alternatively magnetization at slightly different times. Excluding two anomalous directions that cannot be allocated to either polarity, the mean bedding-corrected direction is 316.7°, 6.3° ($\alpha_{95}=14.2°$) for the lower, and 324.7°, 5.3° ($\alpha_{95}=17.8°$) for the higher-temperature group. The results indicate that the basal part of the Roper Group contains polarity reversals.

4. Discussion

4.1. Primary remanence

Primary remanences, while important but not essential for determining the location and shape of pole paths, are critical for magnetostratigraphy. Unfortunately, conclusive evidence that a remanence is primary is rarely available. In the present study, the fold tests suggest that remanence components in several units were acquired before deformation. However, such tests do not establish that the components are original, unless it can be shown that the rocks formed during or immediately before the deformation. The fold tests used here served mainly to identify post-folding overprints and to show if bedding corrections should be applied when calculating pole positions.

However, two lines of evidence support circumstantially the primary origin of some of the remanences: a stratigraphically consistent succession of poles along the trajectory, and the consistency of reversal patterns for two pairs of duplicate sections. Ultimately, however, confirmation has to be sought from reversal patterns and pole positions of coeval units in other parts of Australia and elsewhere.

The above argument for primary origin does

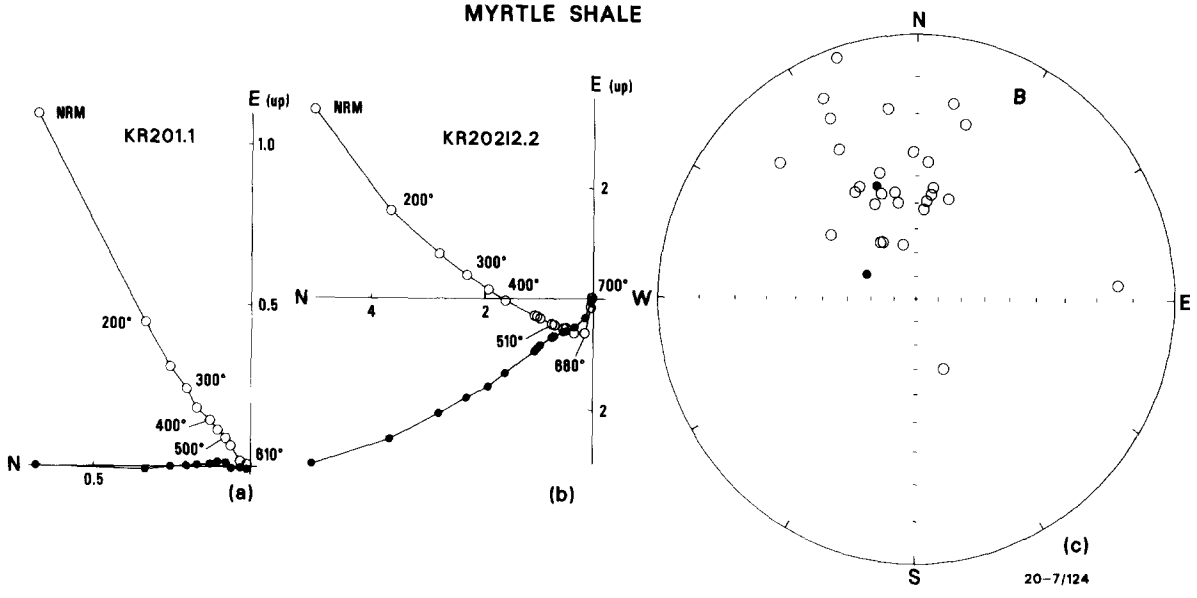


Fig. 19. Myrtle Shale. Orthogonal projection plots show examples of thermal demagnetization giving (a) a single component of northerly negative inclination and (b) a second component of high unblocking temperature and steep positive inclination to the west-northwest. Plot (c) shows remanence directions of the negative-inclination component. Symbols as in Fig. 4.

not apply to remanences of the three igneous formations of the Tawallah Group: the Settlement Creek Volcanics, Hobblechain Rhyolite and Packsaddle Microgranite. These formations lack a fold test, and their poles cluster (although proximity of poles is expected for the rhyolite and microgranite if, as is believed, they are coeval).

The Hobblechain Rhyolite and Packsaddle Microgranite, each with one remanence component that unblocks at ~580°C and another unblocking at ~680°C, have been extensively hematised and K-metasomatized, as have the underlying Gold Creek Volcanics (Knutson et al., 1979). Therefore, the hematite temperature component is most likely secondary. The ~580°C unblocking temperature of the other component suggests that it is carried by magnetite, presumably a remanent of the original mineral. This component is therefore interpreted as primary. (A reliable estimate of its direction is obtained only for the Hobblechain Rhyolite, since the directional scatter in the Packsaddle Microgranite is large.) It should be noted that thin section, XRD and Raman Spectroscopy studies of two typical samples did not reveal magnetite (they showed hematite and, possibly, maghemite); however, remanent magnetite may occur as inclusions and may therefore be too small to be resolved by those methods.

Knutson et al. (1979) attributed the regional metasomatism at Localities 4–6 to hydrothermal fluids related to the formation of numerous breccia pipes. (The fluid temperature of associated carbonate deposition was estimated as 150–300°C, well below the unblocking temperature of the magnetite component.) The minimum age of the pipes is not clear from field relationships. However, Knutson et al. suggested that they formed soon after the extrusion of the volcanics, e.g., as late-stage hydrothermal products of the same magmatic cycle, whereas Wall and Heinrich (1990) suggest that the pipes were produced later during regional tectonism. While the results do not indicate the age of the pipes, the directional similarity of the magnetite and hematite components does favour the suggestion that the regional K-metasomatism occurred soon after the units were emplaced (Knutson et al., 1979).

Where sampled, the Settlement Creek Volcan-

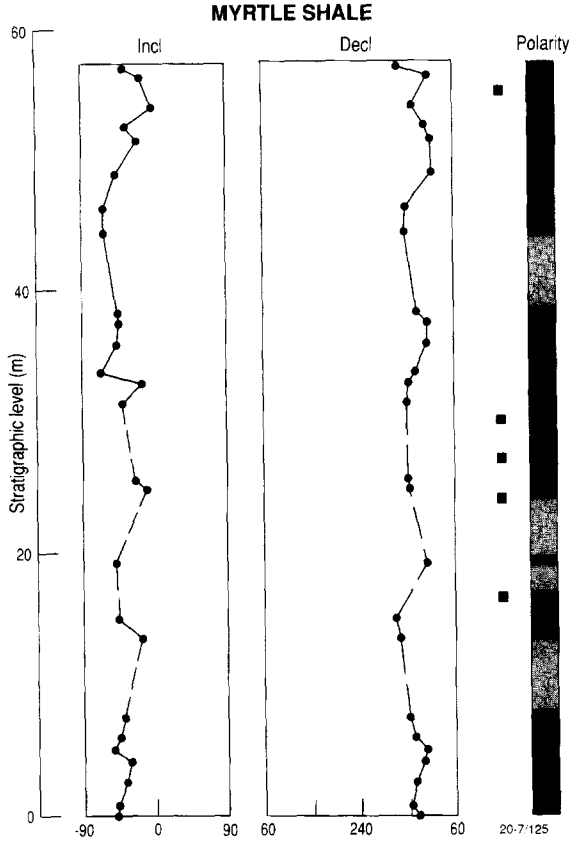


Fig. 20. Reversal column for Myrtle Shale at Locality 12 (type section). Notation as in Fig. 6.

ics are not conspicuously hematized. However, their thin sections show extensive alteration, which Jackson et al. (1987) suggest may be due to K-metasomatism (like the alteration in the nearby younger Gold Creek Volcanics). The remanence is, therefore, interpreted as secondary.

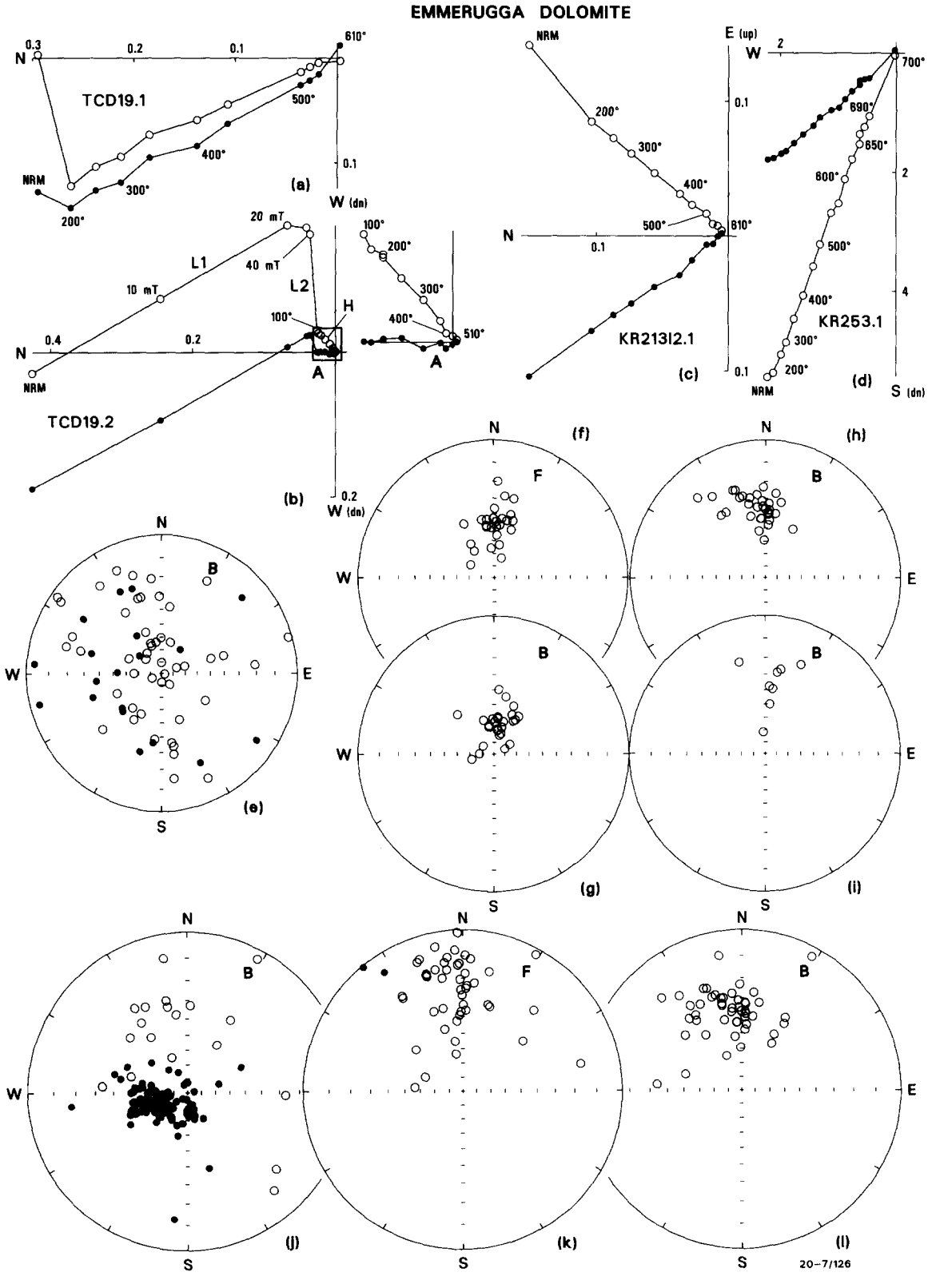
4.2. Overprints

According to the results the McArthur Basin sequences have been overprinted during at least three periods, OP1, OP2, OP3. A possible fourth, OP4, is not well-represented in the data. The earliest overprint, OP1, is related to a metasomatic event, as discussed above. In agreement with field relationships, its remanence direction indicates that the metasomatism was completed before deposition of the McArthur Group because: (1) the cluster of overprint poles lies on the older side of the APWP from the Mallapunyah Formation pole (near the base of the McArthur Group); and (2) the OP1 component was not found in the McArthur Group. The event spanned at least one geomagnetic field reversal.

OP2, the next youngest overprint, has a remanence direction roughly similar to the Late Tertiary geomagnetic field directions. The units affected are the Wollogorang Formation, the top part of Mallapunyah Formation, and the basal part of Amelia Dolomite (at Locality 8). OP2 was not found in the Tooganinine Formation, but its complete absence there is difficult to establish conclusively because of large directional scatter and because, according to the APWP, the overprint and primary remanences would have similar directions. We note that, although OP2 and the Emmerugga Dolomite remanence directions are nearly identical, the latter gave a positive fold test. Significantly, OP2 is absent in the large number of samples from the Balbirini Dolomite, Nathan Group.

The directional similarity of OP2 to the Late Tertiary field appears to be coincidental. The only known Late Tertiary process that may contribute to the remanence in the region is chemical weathering. However, mottling and other weathering indicators are absent at the sites (most Wollogorang Formation samples containing the overprint have no hematite pigment; most Amelia Dolomite samples, although red, are silicified), and the overprint has only one polarity, unlike remanences produced by Tertiary weathering (e.g., Idnurm and Senior, 1978).

The OP2 poles fall on the APWP slightly to the younger side of the Emmerugga Dolomite pole, but appear to be much older than the Balbirini Dolomite pole (Fig. 26). The OP2 group marks the apex of a hairpin bend in the path. The grouping at the apex, relative timing on the path, and absence of OP2 in the Balbirini Dolomite are consistent with overprinting during a likely period of very active rifting and associated high heat flow, and of pervasive alteration during deposition of the uppermost McArthur Group, comprising the Barney Creek Formation and Batten Subgroup. Each of the overprinted sequences is


EMMERUGGA DOLOMITE
TCD19.1
TCD19.2
KR2131.2.1
KR253.1
NRM
L1
L2
H
A
(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
(i)
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(k)
(l)
W (dn)
E (up)
S (dn)
20-7/126

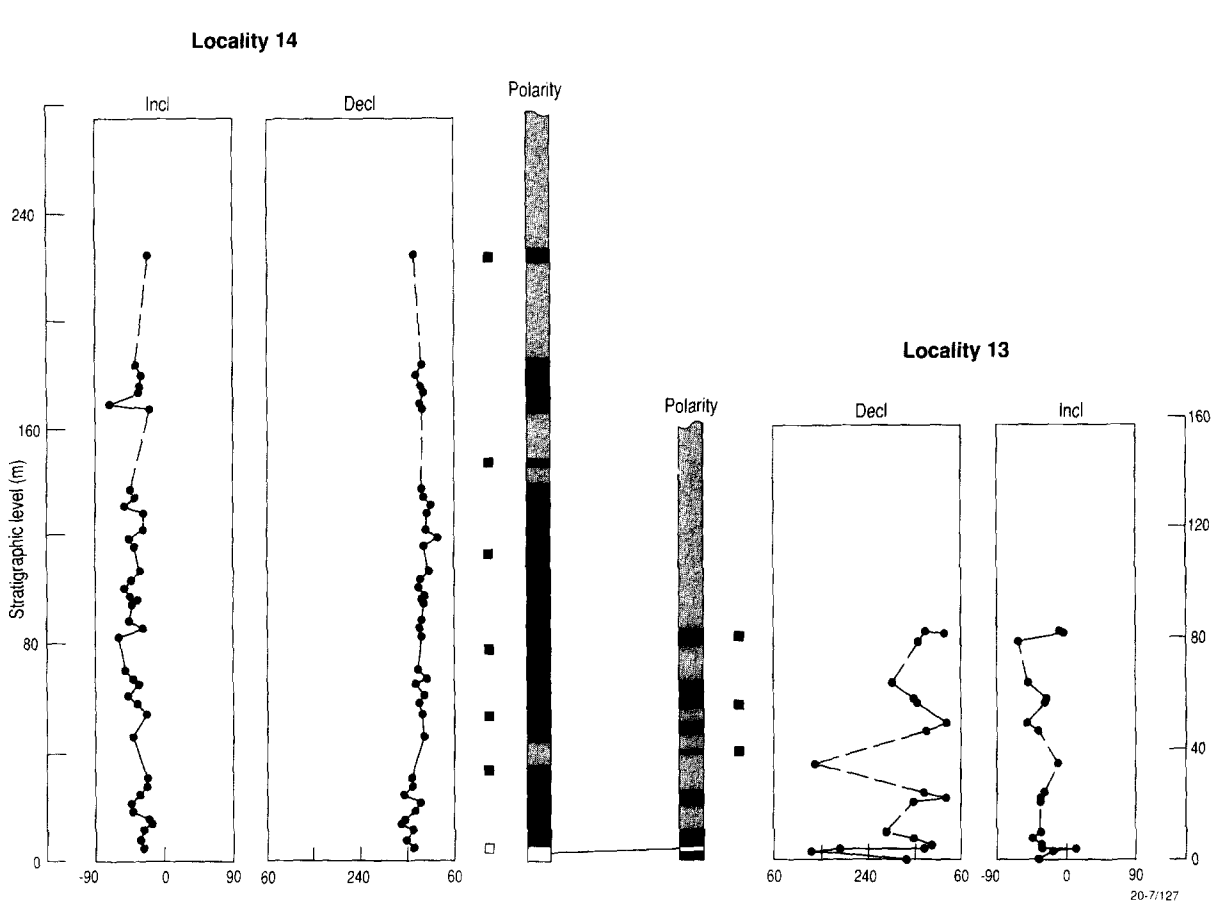


Fig. 22. Reversal columns for Emmerugga Dolomite at Locality 14 (type section) and Locality 13. Notation as in Fig. 6.

near a prominent north- or northwest-trending fault which may have channelled heat flow by fluid convection. Indeed, a convective system, driven by radiogenic heat from basement granites, has been proposed for the Pb–Zn deposits of the McArthur Basin (HYC) and Mount Isa Inlier by Solomon and Heinrich (1992), who suggest that these large ore bodies may have been produced over a period of 0.1–1.0 m.y. by episodic fluid circulation at temperatures of ~230°C.

OP3, the third youngest overprint, is characterised by steep, positive inclinations to the west-southwest, and was found in the three principal

Fig. 21. Emmerugga Dolomite. Orthogonal projection plots (a) and (b) illustrate the inability of thermal demagnetization alone to separate remanence components at the type section: (a) shows a pseudocomponent obtained from TH demagnetization; (b) shows the separation of the remanence into three components (*L1*, *L2* and *H*) in a second specimen from the same sample by partial AF demagnetization, followed by TH demagnetization. Other orthogonal projection plots give examples of TH demagnetization at Locality 13, which has (c) a component with negative inclination similar to the H component at Locality 14 and (d) a steep, positive-inclination component interpreted as an overprint. Stereo plots for Locality 14 show (e) randomness of the pseudocomponent directions obtained from TH demagnetization; (f) and (g) the L2 component before and after bedding correction, obtained by AF/TH demagnetization; (h) and (i) high unblocking temperature component in the lower (h) and higher (i, recrystallised) part of Locality 14. Plot (j) shows remanence directions from Locality 13; (k) and (l) show combined hard component directions with negative inclinations from the two sampling localities before (*F*) and after (*B*) bedding corrections. Symbols as in Fig. 4.

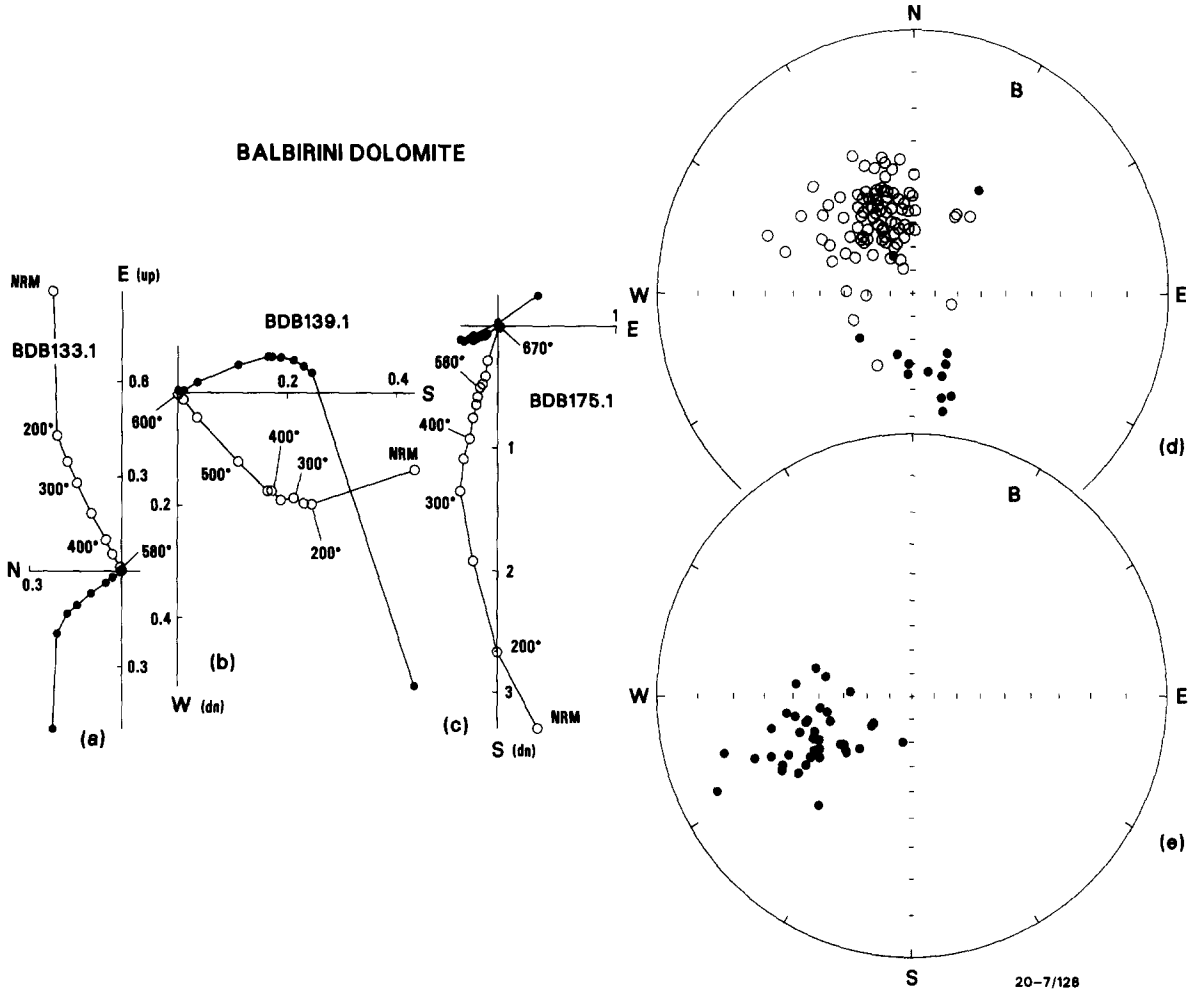


Fig. 23. Balbirini Dolomite. Examples of orthogonal projection plots showing components with (a) negative NW inclination, (b) positive SE inclination and (c) positive WSW inclination. Stereo plots show remanence directions interpreted as (a) primary and (b) overprint. Symbols as in Fig. 4.

dolomitic units: the Amelia, Emmerugga and Balbirini Dolomites. It may also be present as a minor component in the Myrtle Shale. OP3 is especially strong in the Emmerugga Dolomite at Locality 13, where it was previously interpreted as primary (Idnurm and Giddings, 1988). A similar overprint occurs in the Kombolgie Formation (Giddings and Idnurm, 1993) at the northwestern part of the basin. Although fold tests are not available, an overprint origin of the remanences seems clear: the poles plot well out of sequence and group markedly better when not corrected for bedding. Moreover, for the Amelia and Emmerugga Dolomites OP3 is associated with intense reddening, and for the Balbirini Dolomite OP3 is found only in the recrystallized upper part. The overprinting seems to be confined to specific locations: it is absent in the Emmerugga Dolomite type section, but occurs in the Balbirini Dolomite 4 km to the north. Although OP3 has no known minimum age, its poles plot on the 1600–1500 Ma segment of the Australian APWP and are inconsistent with younger path

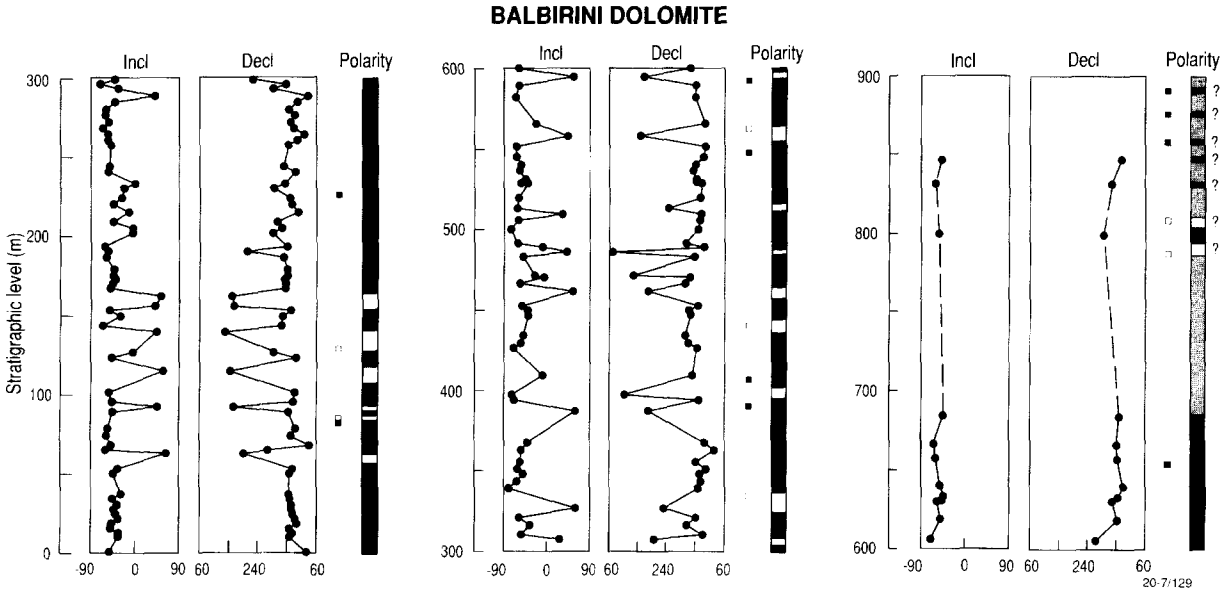


Fig. 24. Reversal column for Balbirini Dolomite at Locality 16 (type section). Polarity identifications in the upper 200 m section affected by extensive recrystallization are tentative. Notation as in Fig. 6.

segments, including the Palaeozoic. The cause of the remagnetization is uncertain.

The pole of the fourth overprint, OP4, differs from OP2 by only 13°, and the two overprints may therefore be the same. On the other hand, OP4 is also close to the 1550–1500 Ma path segment and is associated with fine-scale ferruginous mottling not observed in the other units. Therefore, the pole may represent a different overprint: the ferruginization may be due, for example, to an episode of fluid circulation along an ancient aquifer system at 1550–1500 Ma. We note that, as for OP2, the remanence direction of OP4 resembles the present dipole field direction. The ferruginization is, however, unlikely to be recent: parts of the ferruginous zone are silicified and therefore of low permeability. The mottling is controlled by bedding planes (not rock joints and fractures) and is related to stratigraphy rather than to present topography.

4.3. Assignment of bedding corrections to overprints

Table 2 shows by italics if the field- or bedding-corrected coordinate is appropriate. While for some poles the choice is clear, for others it represents a compromise, as neither alternative is likely to be completely correct.

The poles nominated in Table 2 for the oldest overprint, OP1, are bedding-corrected: the overprinting seems to have shortly followed the formation of the host rocks before significant tectonism. Bedding corrections are also applied to the three poles of the second overprint, OP2. These lie close to the Emmerugga Dolomite primary pole and their remanence was therefore probably acquired during deposition of the Umbolooga Subgroup. Since the overprinted Mallapunyah Formation and Amelia Dolomite belong to the same subgroup, deposition of the host sequences and their overprinting were not separated by regional unconformities. The primary and overprint components in these two formations should consequently have a common history of regional tilting, unless the units were deformed locally (e.g., by faulting) in the intervening period. There is no evidence of such deformation at the two sampling sites. For the third unit that contains OP2, the Wollogorang Formation, the overprint post-dates localised deformation, which we assume occurred at the

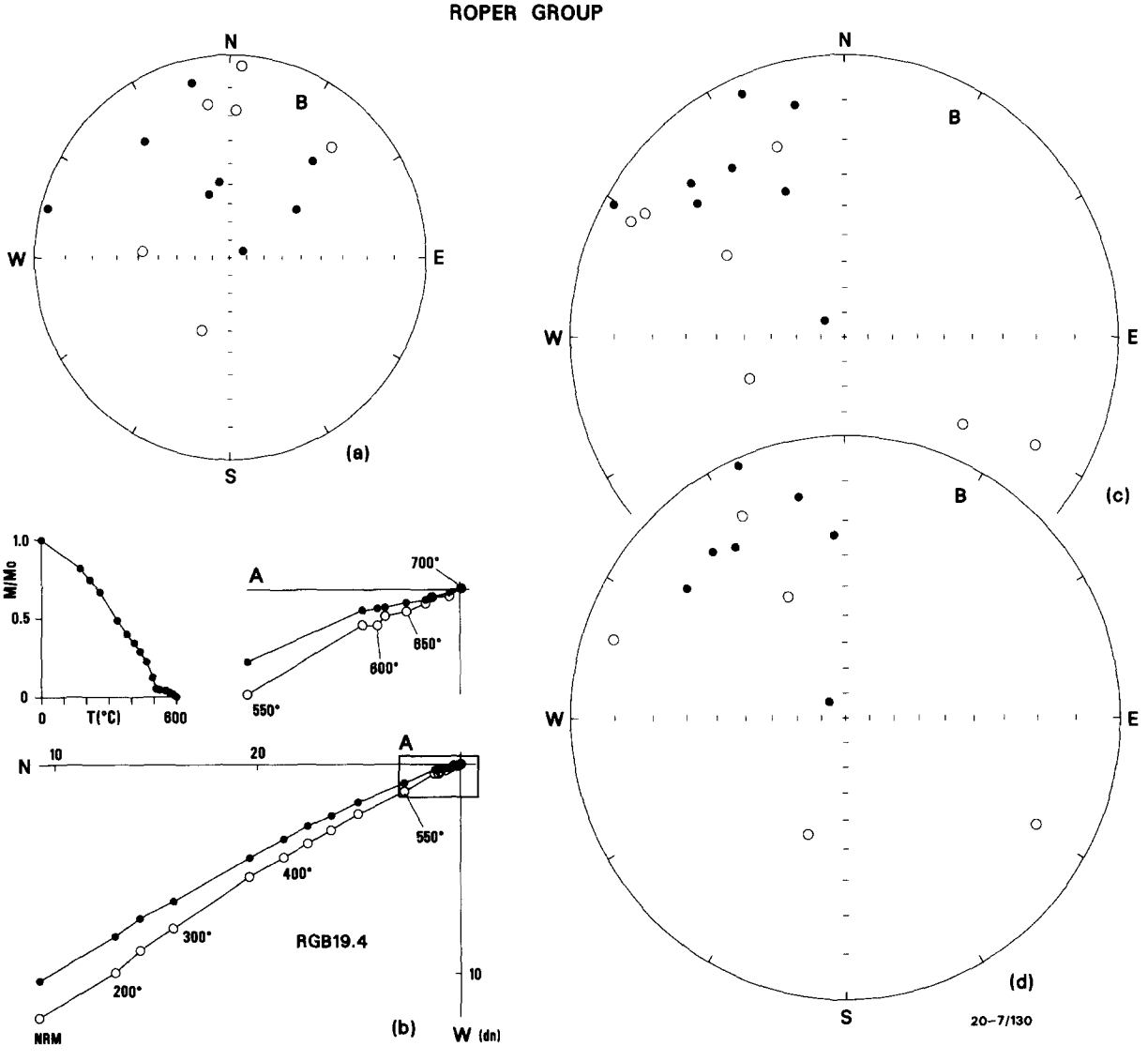


Fig. 25. Roper Group. Plots show (a) highly scattered remanence directions from pigmented samples, (c) intermediate temperature component in non-pigmented samples, and (d) high-temperature component. Diagram (b) shows an example of thermal demagnetization of non-pigmented samples: characteristically, the elimination of the intermediate component is marked by an inflection in the normalised intensity plot. Symbols as in Fig. 4.

time of the Tawallah–McArthur Group unconformity or during block faulting when the Umbolooga Subgroup was deposited (see Sect. 3.8). In either case, the bedding dips of the nearby Masterton Sandstone would give the appropriate tilt correction.

Reliable bedding corrections are unavailable for OP3 and OP4, which are older than the Roper Group. Both post-date the Nathan Group, and therefore also the deformations associated with the three earlier regional unconformities. Uplift, deformation and tilting of fault-blocks continued after OP3 and OP4, as seen for example in the Roper Group at the Abner Range. It is

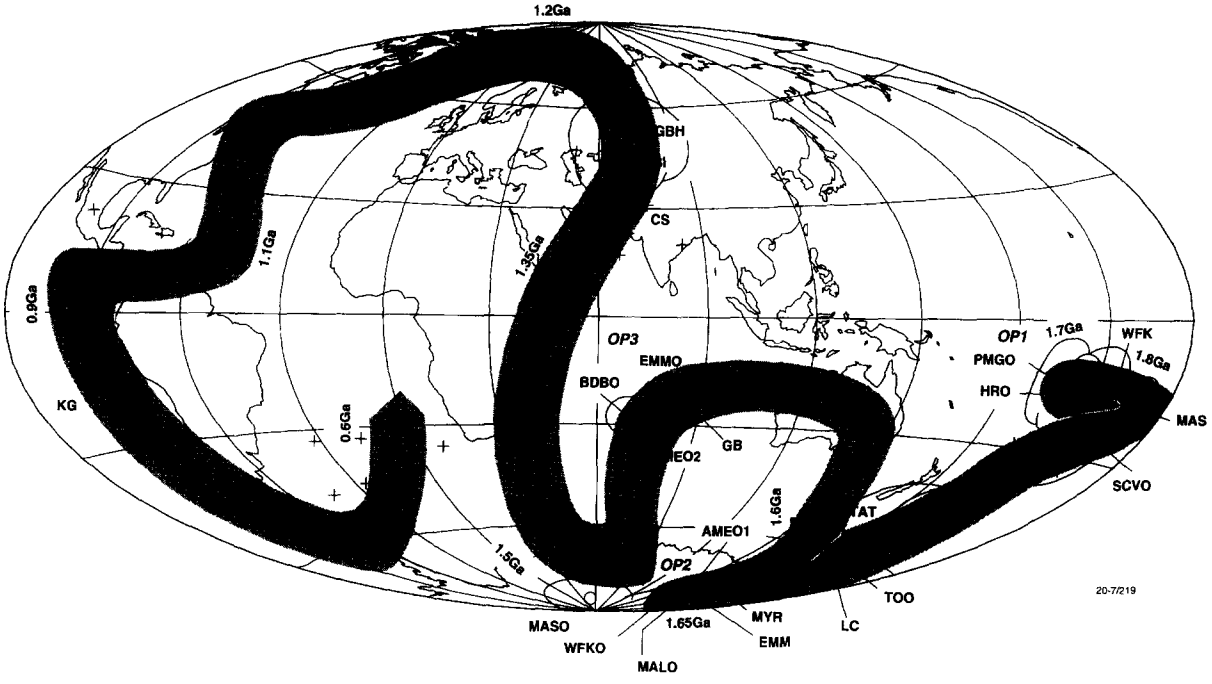


Fig. 26. Poles from the present study plotted on the 1800–600 Ma apparent polar wander path for Australia (after Idnurm and Giddings, 1988). The new poles, shown with 95% confidence limits, are listed in Table 2. New poles interpreted as primary are shown by dots, overprint poles by circles and poles from previous studies by crosses. *OP1*, *OP2*, *OP3* and *OP4?* indicate overprint pole groupings. The figure is adapted from Idnurm and Giddings (1988) which lists the previous poles (previous preliminary McArthur Basin poles and the pole from the Giles Complex has been omitted, see Camacho et al., 1991). Poles reported more recently are from the Kulgera Dyke Swarm (*KG*) (Camacho et al., 1991) and from Mount Isa Inlier (*IM* and *IAR*) (Tanaka and Idnurm, 1994). Non-McArthur Basin poles mentioned in the text are from the Lunch Creek Gabbro (*LC*), Gawler Block B dykes (*GB*), Morawa Lavas (*ML*) and Campbell Sandstone (*CS*).

not clear, however, how localized the subsequent tilting was and, therefore, what proportion of the total tilt should be used for the corrections. Because the three OP3 poles become appreciably more scattered when corrected for the observed tilts, we consider the field-corrected poles to be the more accurate ones. The neglect of post-overprinting tilts would cause errors, which may be reflected in the small differences between OP3 directions in the Amelia, Emmerugga and Balbirini Dolomites: Watson's *F* test (Watson, 1956) shows that the differences are significant at the 99% confidence level.

4.4. Apparent polar wander path

Fig. 26 shows the 22 poles of the present study superimposed on the Australian APWP for the period 1800–600 Ma (Idnurm and Giddings, 1988), and Fig. 27 shows a detail of the pole se-

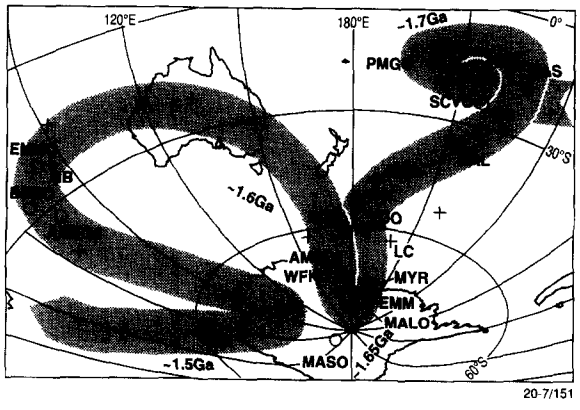


Fig. 27. Detailed APWP for the Tawallah, McArthur and Nathan Groups. For description see Fig. 26.

quence for the lower three stratigraphic groups. Twelve of the new poles are interpreted as primary, ten as secondary. As discussed in the previous section, the latter belong mainly to three periods of overprinting.

The poles and the path segment they define update the preliminary results of Idnurm and Giddings (1988). Although the new pole positions differ variously from those reported earlier (which were based on about 5% of the final data), the trajectory itself needs little change. The most important revision is the reinterpretation of the preliminary Emmerugga Dolomite pole, which is now identified as an overprint (EMMO). Also revised are the previous palaeolatitude estimates according to which some members of the McArthur Group were high-latitude deposits. All sequences, including those of the Roper Group, now appear to have been deposited in low to moderate latitudes (Fig. 28, Table 2), consistent with a predominantly carbonate lithology and an abundance of stromatolites, evaporites and red beds. The update restores the correct stratigraphic order for the Amelia/Tatoola and Tooganinine poles.

The Tawallah–McArthur–Nathan APWP covers an angular distance of about 120°. Geochronological dates from two units, the Tanumbirini Rhyolite at the top of the Tawallah Group, and the Barney Creek Formation in the middle of the McArthur Group (Fig. 3), give this part of the sequence a minimum age span of ~75 m.y.; the actual span may exceed 100 m.y. Although the Roper Group poles are included in Fig. 26, they are tentative, giving only a rough indication of where the path segment is located for the lower formations. The primary nature of these poles needs confirmation by the fold test, or other evidence.

Several Proterozoic poles from other tectonic units of Australia overlap the McArthur Basin poles. Pole LC from the Lunch Creek Gabbro, Mt Isa Inlier, falls on the 1700–1650 Ma part of the APWP (Fig. 26). Pole GB from dolerite dykes of the Gawler Craton overlaps OP3. Preliminary poles from the Morawa Lavas (ML) and the overlying Campbell Sandstone (CS) in the Yilgarn Block lie close to the Roper Group poles.

The overlaps are consistent with the assembly of the Australian Craton in the Precambrian, but the age constraints for the remanences are too poor for conclusive tests. Because the Lunch Creek Gabbro is metamorphosed, its U–Pb zircon age of 1740±25 Ma (Page, 1983) represents only the maximum age for the pole LC. Again, the GB dykes have a rough preliminary Rb–Sr date of 1660±100 Ma (W. Compston and A.R. Crawford in Giddings and Embleton, 1976) which, together with a more reliable Rb–Sr minimum age of 1549±63 Ma for dykes of the same region (Mortimer et al., 1988), is merely consistent with a younger than 1600 Ma age for the overprint OP3. The Morawa Lavas have a Rb–Sr date of 1360±140 Ma (Compston and Arriens, 1968; precision P.A. Arriens, pers. commun., 1974), also only broadly consistent with the Roper Group minimum ages of 1429±31 Ma and 1510 Ma (see Section 1.2). The pole ages there-

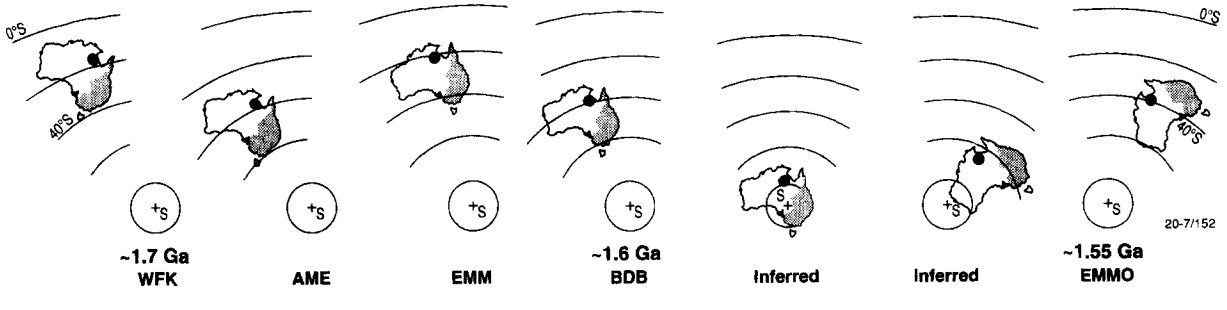


Fig. 28. Palaeolatitudinal change of the McArthur Basin (dot) during the period ~1700 to ~1550 Ma. The Phanerozoic Tasman Fold Belt is shown shaded.

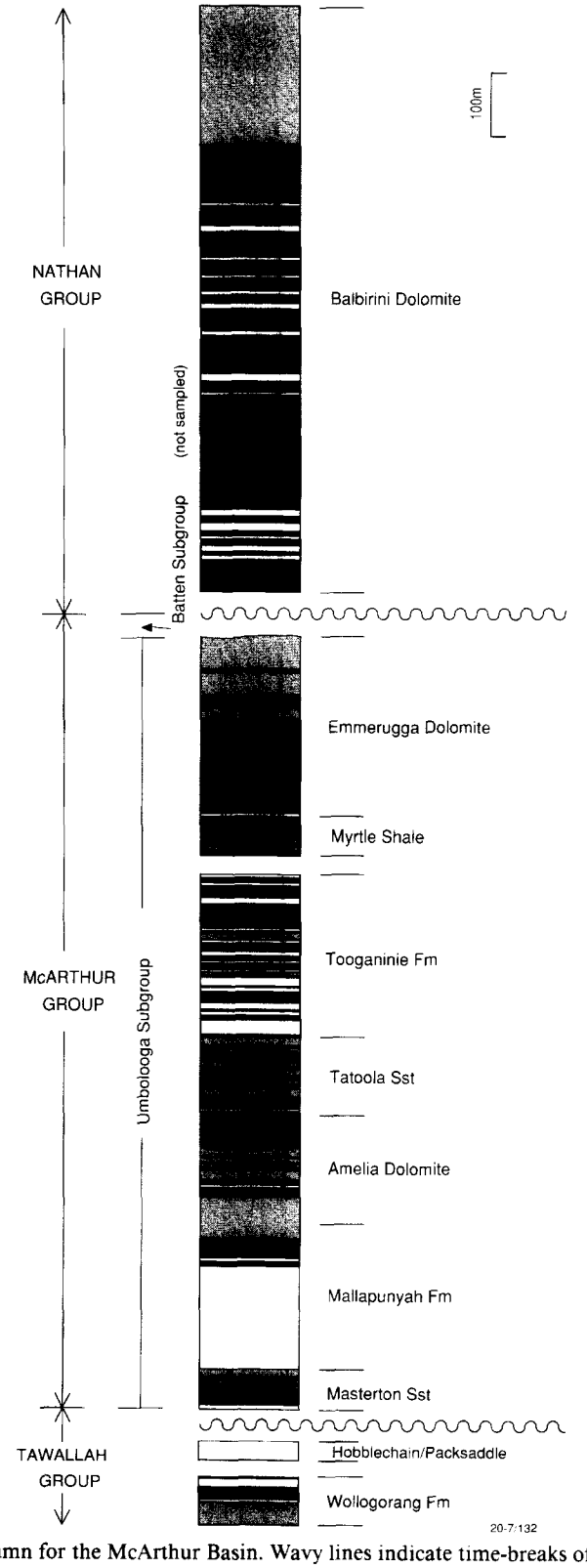


Fig. 29. Composite reversal column for the McArthur Basin. Wavy lines indicate time-breaks of possibly tens of million years. Notation as in Fig. 6.

fore need to be far better defined to test the congruity of the continental assemblage.

4.5. Magnetostratigraphy

The composite reversal column for the McArthur Basin is shown in Fig. 29. The column represents an accumulated stratigraphic thickness of 2100 m, but includes many gaps where the polarity is undetermined because of unstable remanence, overprinting, unsampled parts of the sequence, and sections missing because of regional unconformities. Non-uniform rates of sedimentation, especially short time-breaks, would distort further the relative spacing of the reversals. Nevertheless, some distinctive characteristics of the reversal pattern that are useful for correlation and dating are evident in Fig. 29.

The lower part of the column, below the base of the Tooganinine Formation, is characterised by infrequency of reversals. That part contains major gaps due to destruction of the primary remanence and the presence of a regional unconformity between the Tawallah and McArthur Groups. The duration of the latter is uncertain, but could be several tens of million years. The Mallapunyah Formation with its remanence directions of predominantly negative inclination contrasts the mostly positive-inclination remanences of the other formations. Reversal frequency increases in the lower half of Tooganinine Formation, but is reduced again in the upper half, and gives way to a constancy of polarity in the overlying Myrtle Shale and Emmerugga Dolomite. Reversals reappear in the Balbirini Dolomite, which is separated by another time gap of possibly tens of million years from the underlying parts of the column.

The consistency of the reversal data was tested by duplicate sampling at different localities of the same stratigraphic interval, one in the Mallapunyah Formation and another in the Emmerugga Dolomite. Duplicate sections of the Mallapunyah Formation have similar patterns (Fig. 12), except for an extra reversed polarity interval at Locality 9. The latter is based on only one sample and, because of its narrowness, could have been missed at the first locality. The otherwise single-polarity Emmerugga Dolomite contains a narrow reversed interval near its base at both localities (Fig. 22).

The duplicate sections have too few reversals to provide a stringent test. Moreover, the sections for the Mallapunyah Formation are separated by only 4 km and those for the Emmerugga Dolomite by 30 km; therefore, their remanences could be affected by the same regional remagnetization processes. Further confirmation is needed from more widely spaced sequences. However, the preliminary results are encouraging.

Several basins in northern Australia contain sedimentary sequences of an age similar to those of the McArthur Basin (Plumb, 1985, 1990). The present results suggest that detailed correlations between some members of these sequences may be feasible by the use of distinctive polarity reversal patterns. The method would have to be used in conjunction with other evidence to confirm broad similarity of ages: isotopic dates, pole positions, etc.

5. Conclusions

Twelve poles obtained from the southeastern McArthur Basin are interpreted as primary, including two preliminary poles. The ten better-established poles define a path segment for Australia of about 120° angular length which spans a Palaeo–Mesoproterozoic interval estimated as at least 75 m.y. long and possibly much more. They increase by about 30% the Proterozoic palaeomagnetic database for Australia and precisely define a segment of the path that was previously constrained by only one pole. The poles suggest that during deposition of the Tawallah, McArthur and Nathan Groups the region was in low to middle latitudes. Similar palaeolatitudes are tentatively obtained for the much younger Roper Group.

Another ten poles are interpreted from fold tests and other evidence as secondary. These record three or possibly four periods of overprinting. The first, related to metasomatism, occurred before or at the beginning of deposition of the

McArthur Group. The second, related possibly to hydrothermal activity and rifting, occurred later in the McArthur Group deposition, and may have been coeval with the deposition of the Barney Creek Formation hosting the HYC stratiform Pb–Zn–Ag deposit. The third overprint selectively affected dolomitic units, and appears to post-date the Nathan Group by several tens of million years. The possible fourth overprint, from immediately below the Tawallah/McArthur Group unconformity, appears to be associated with iron deposition in an ancient aquifer. The sequences also contain recent overprints which appear to be due to lightning strikes and low-temperature remagnetization of goethite.

The remanences were measured through a discontinuous stratigraphic interval of which they represent 2100 m. A reversal column was determined for the sequence, and two duplicate sections gave consistent patterns. The column contains many gaps where the units are poorly exposed or where overprinting has been too severe to recover the primary remanence. Nevertheless, the reversal pattern appears to be sufficiently distinctive for use as a tool in correlation of rock sequences, provided this is in conjunction with other data such as pole positions or U–Pb zircon ages. This is only the second magnetostratigraphic study on Palaeoproterozoic or older sequences (Bingham and Evans, 1976).

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